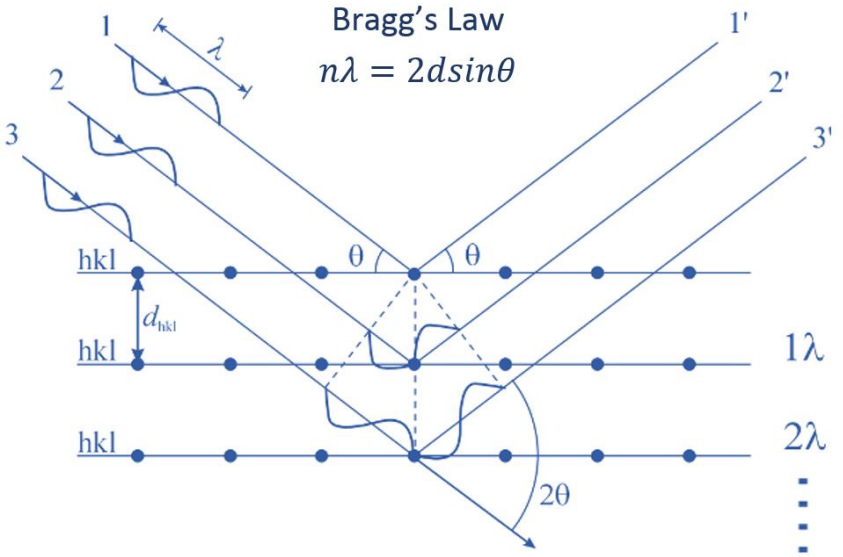
Fundamental Physics II
Optics - Chapter 2

For B1 Students
Academic year 2024-2025



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OVERVIEW

Chapter	Contents	Hours			Ref./Resources	Assignment(s)
		Lect.	Exr.	Prc.		
1	The nature and propagation of light	2.0	1.0			Exercises
2	Geometric optics	5.0	2.0			Exercises
3	Interference Midterm test (0.5h)	3.0	1.5			Exercises
4	Diffraction	3.0	1.5			Exercises
5	Photons: Light waves behaving as particles*					Exercises
		13	6			

Assessment/ Evaluation		
	Attendance/Attitude	**10%**
	Exercise(s)	0%
	Practicals	0%
	Mid-term test	**30 %**
	Final exam	**60 %**

Reference Literature:

[1] Young and Freedman – University Physics with Modern Physics 15th Edition (2020)

Chapter 2. Geometric optics (5.0h)

2.1 Reflection and Refraction at a plane surface

2.2 Reflection at a spherical surface

2.3 Refraction at a spherical surface

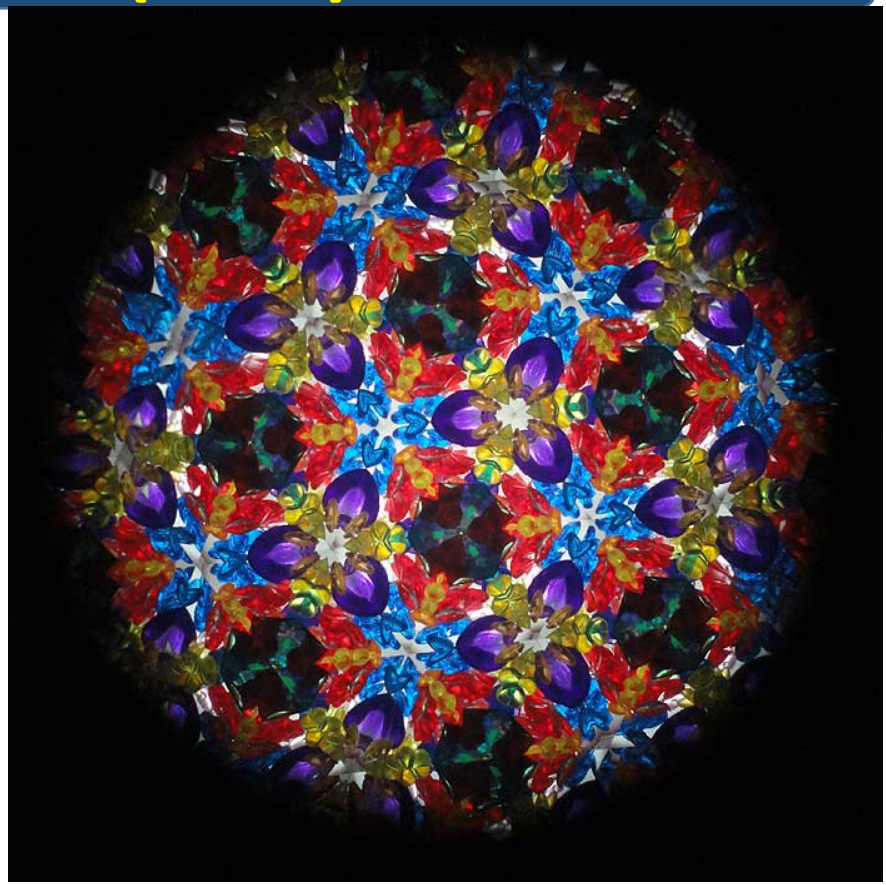
2.4 Thin lenses

2.5 Cameras

2.6 The eye

2.7 The magnifier

2.8 Microscopes and Telescopes



Kaleidoscope

List of Exercises:

In textbook “University Physics with Modern Physics” 15th Edition, Hugh D. Young, Roger A. Freedman, A. Lewis Ford (2020)

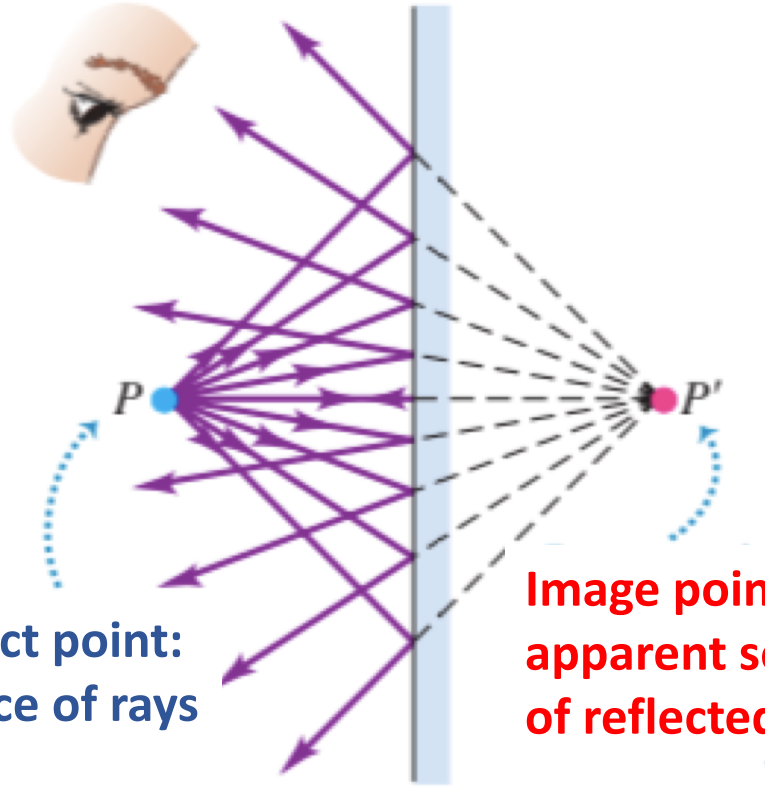
34. 3, 5, 8, 13, 19, 21, 26, 29, 31, 32, 40, 43, 45, 50, 52, 54, 58, 59, 62, 63, 66, 71

Chapter 2. Geometric optics (5.0h)

2.1 Reflection and Refraction at a plane surface

What is an object?

- Object: Anything from which light rays radiate.
  - ✓ Light could be emitted by the object itself if it is *self-luminous*.
  - ✓ Light could be emitted by another source and then reflected from the object.



Object point: source of rays

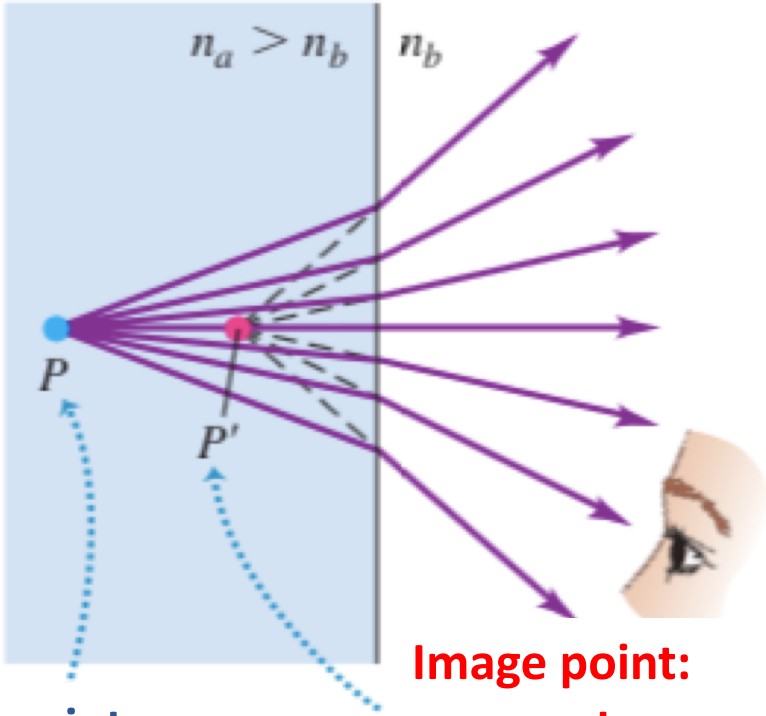
Image point: apparent source of reflected rays

Plane mirror

Light rays from the object at point *P* are reflected from a plane mirror.

$n_a > n_b$
=> *P'* is closer to the surface than *P*

The refracted rays entering the eye look as though they had come from image point *P'*.



Object point: source of rays

Image point: apparent source of refracted rays

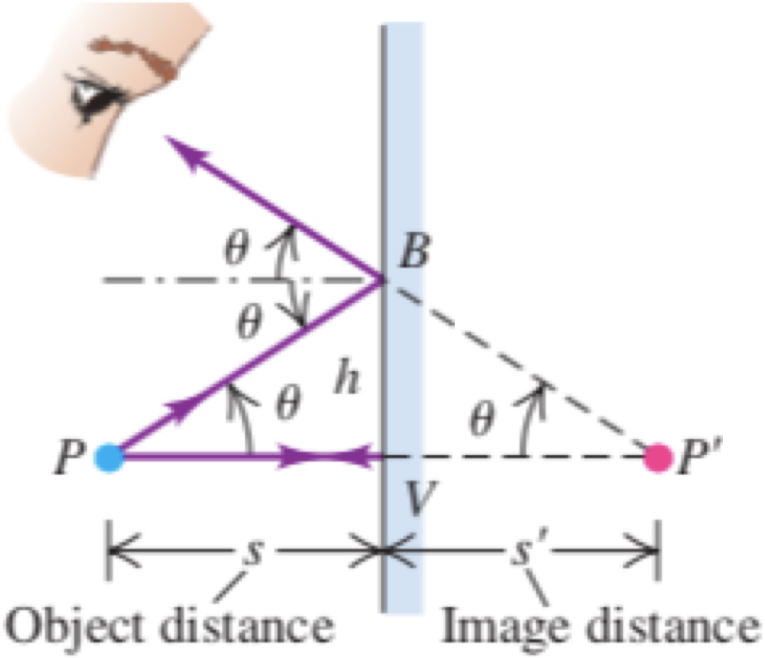
Light rays from the object at point *P* are refracted from a plane interface.

Chapter 2. Geometric optics (5.0h)

2.1 Reflection and Refraction at a plane surface

2.1.1 Image formation by a plane mirror

The location of the virtual image *P'* of object *P* formed by a plane mirror?



Two incoming rays:

Two reflected rays:

- *PV* is incident normally on the mirror => returns along its original path
- *PB* is incident at angle θ with *PV* => reflects at an equal angle with the normal

} Image *P'*

s = *PV*: object distance

s' = *P'V*: image distance

$|s'| = |s|$

Chapter 2. Geometric optics (5.0h)

2.1 Reflection and Refraction at a plane surface

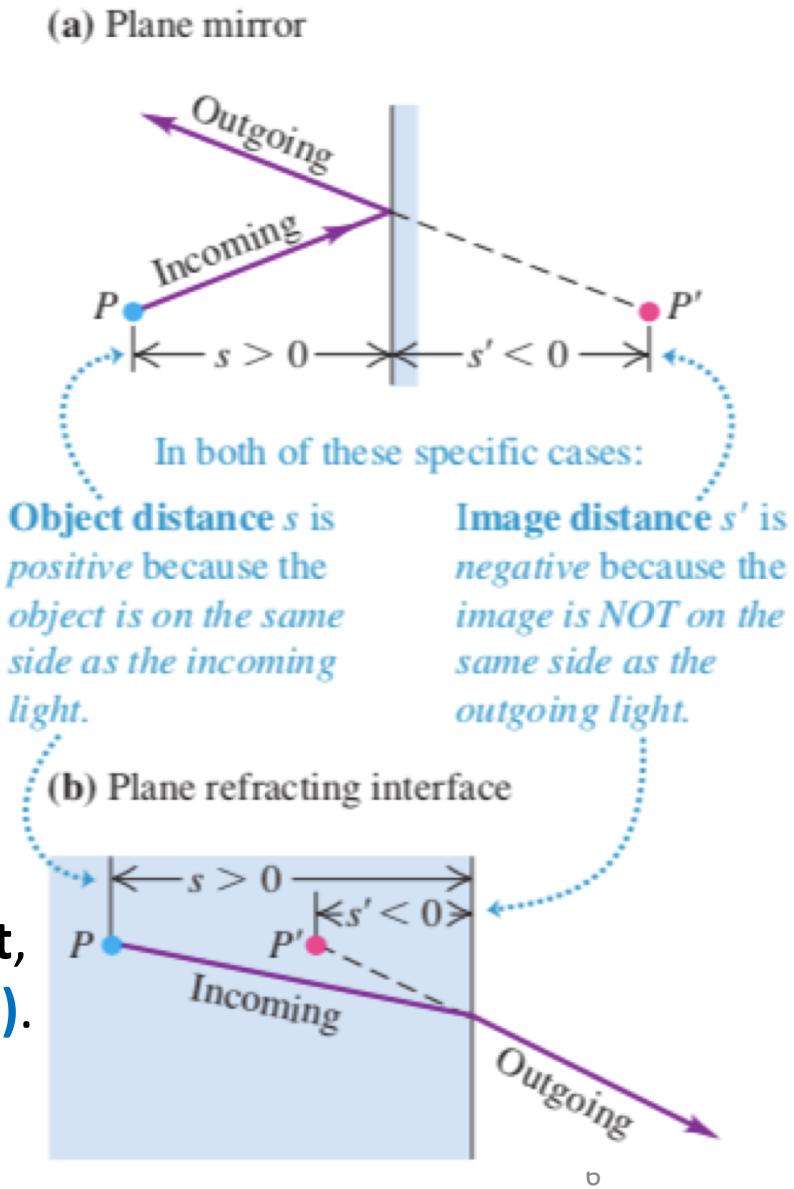
2.1.2 Sign rules

1. Sign rule for the **object distance:**
When the **object** is on the **same side** of the **reflecting or refracting surface** as the **incoming light**,
the object distance *s* is **positive**; otherwise, it is **negative**.

2. Sign rule for the **image distance:**
When the **image** is on the **same side** of the **reflecting or refracting surface** as the **outgoing light**,
the image distance *s'* is **positive**; otherwise, it is **negative**.

3. Sign rule for the **radius of curvature of a spherical surface:**
When the **center of curvature *C*** is on the **same side** as the **outgoing light**, the **radius of curvature** is **positive (R > 0)**; otherwise, it is **negative (R < 0)**.

For plane mirror: $s' = -s$



Chapter 2. Geometric optics (5.0h)

2.1 Reflection and Refraction at a plane surface

2.1.3 Image of an extended object: plane mirror

- Height of an object/image: the distance from the head to the tail of an arrow.
- The **lateral magnification** *m*:

$$m = \frac{y'}{y}$$

- The image is **erect**:
  the image arrow points in the same direction as the object arrow.

$$m > 0$$

- The image is **inverted**:
  the image arrow points in the direction opposite to that of the object arrow.

$$m < 0$$

- An image formed by one surface or optical device can serve as the object for a second surface or device.

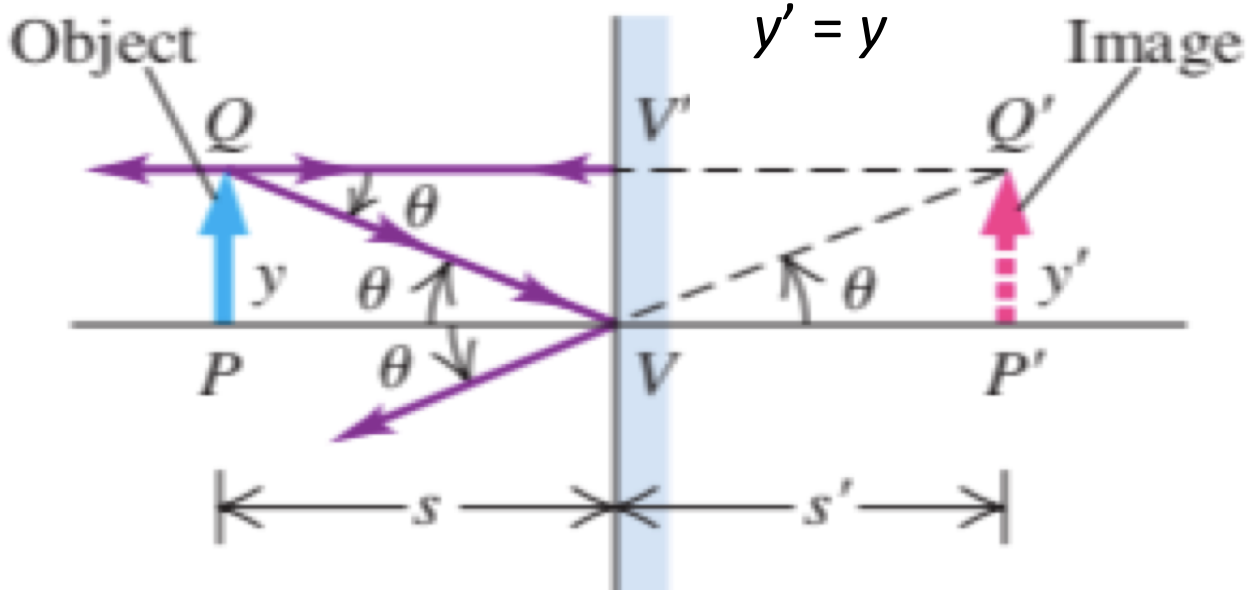
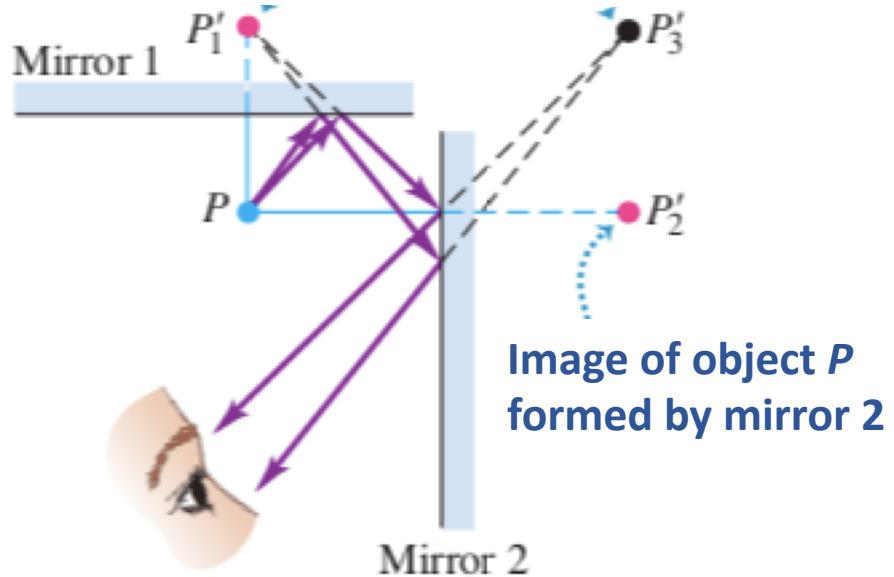


Image of object *P* formed by mirror 1

Image of image P_1' formed by mirror 2



Chapter 2. Geometric optics (5.0h)

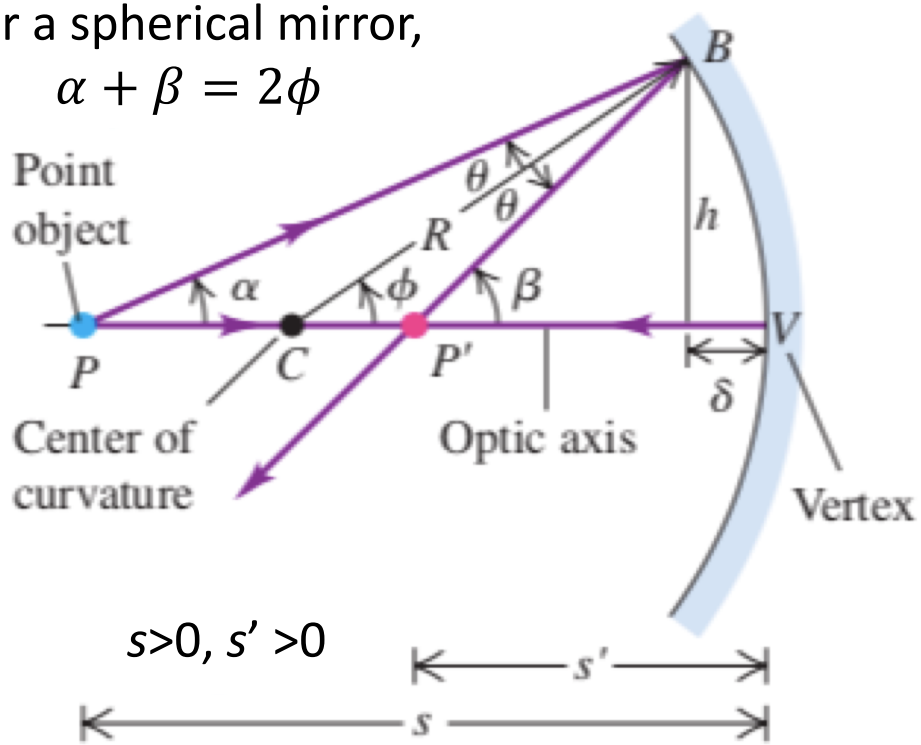
2.2 Reflection at a spherical surface

2.2.1 Image of a point object: spherical mirror

- A spherical mirror with radius of curvature *R*, with its concave side facing the incident light.

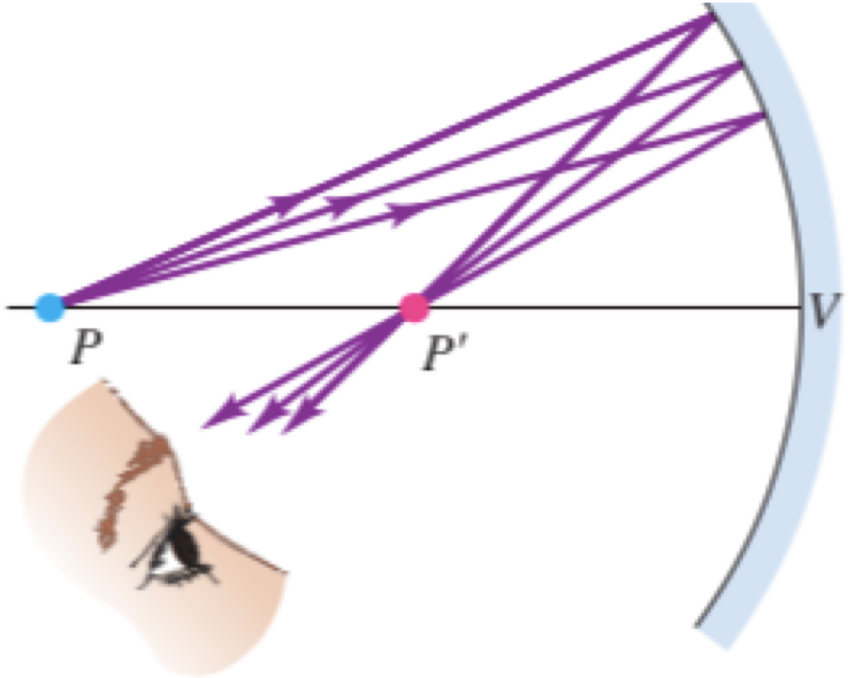
For a spherical mirror,

$$\alpha + \beta = 2\phi$$



- The **center of curvature** of the surface is at *C*.
- The **vertex** of the mirror is at *V*.
- *CV* is called the **optic axis.**

All rays from *P* that have a small angle α pass through *P'*, forming a real image.



- *PV* passes through *C*, strikes the mirror normally
  => is reflected back on itself
- *PB* at angle α with the axis, strikes the mirror at *B*
  => is reflected: $\theta' = \theta$

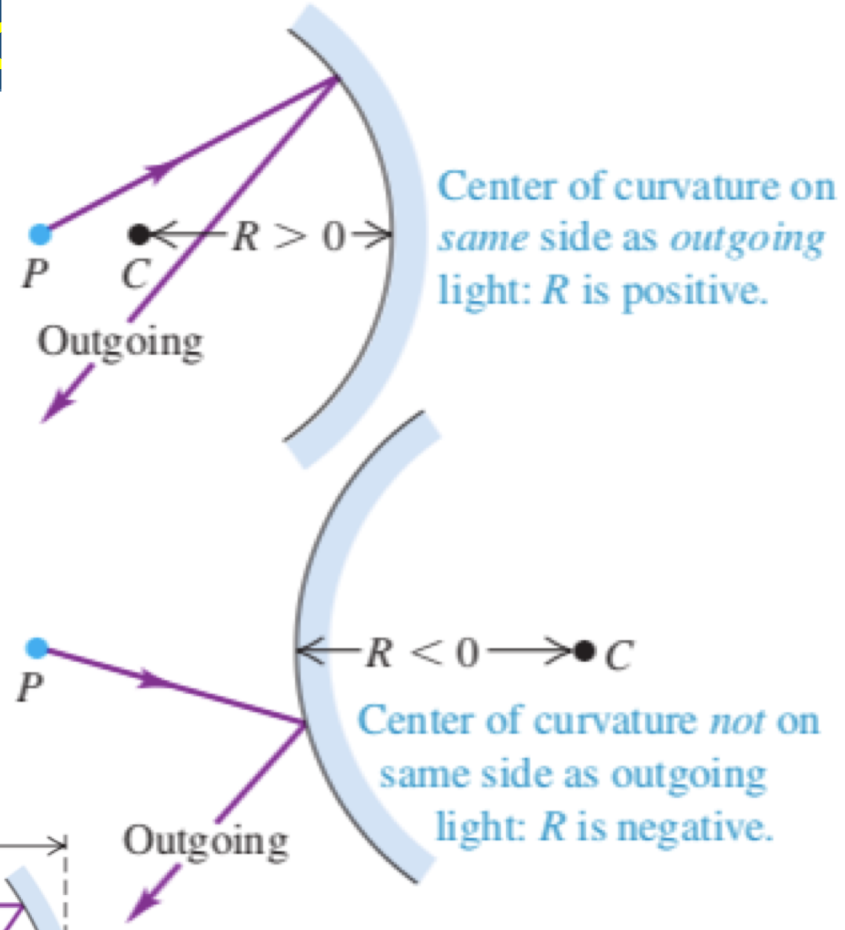
} Image *P'*

2.2 Reflection at a spherical surface

2.2.1 Image of a point object: spherical mirror

$$\alpha + \beta = 2\phi$$

$$\frac{1}{s} + \frac{1}{s'} = \frac{2}{R}$$ object– image relationship



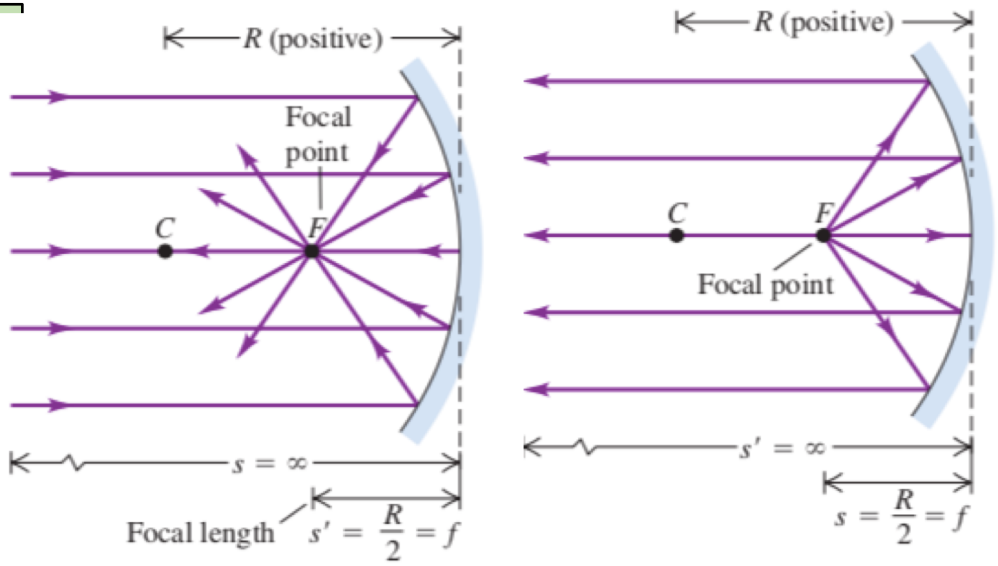
- **Paraxial rays**: the rays are nearly parallel to the axis and close to it.
- **Spherical aberration**: the image is ”smeared out” when the α is large.

2.2.2 Focal point and focal length

$$f = \frac{R}{2}$$ Focal length

$$\frac{1}{s} + \frac{1}{s'} = \frac{1}{f}$$

object– image relationship



Focal point *F*

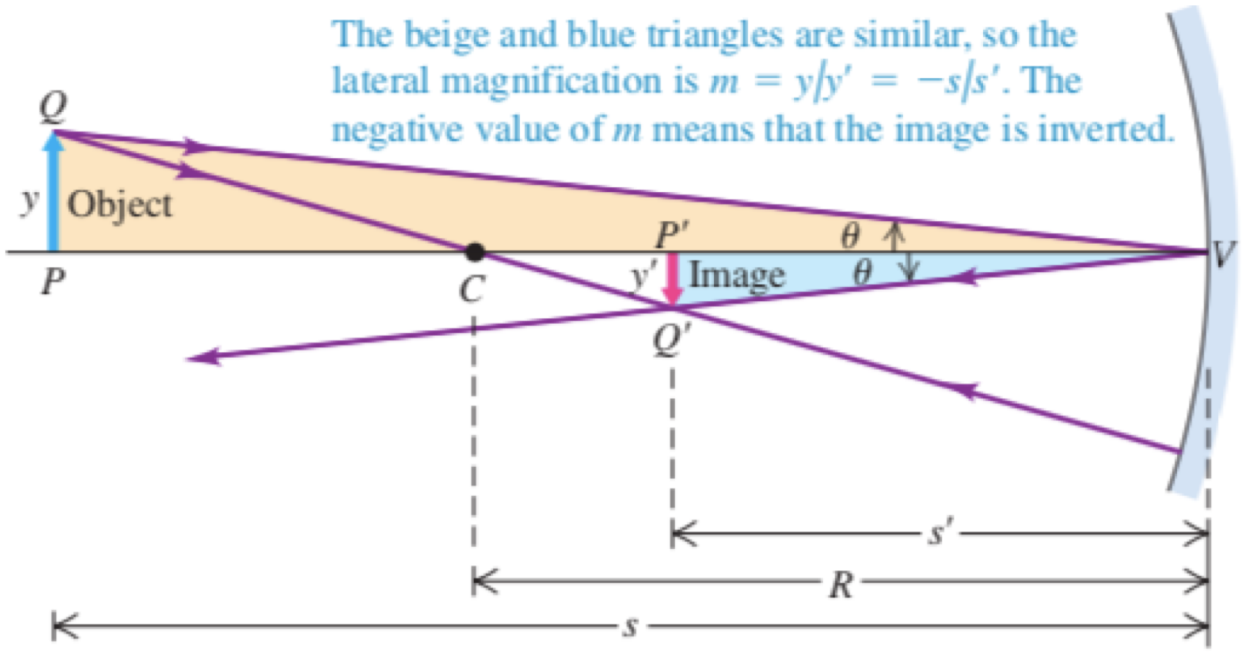
Chapter 2. Geometric optics (5.0h)

2.2 Reflection at a spherical surface

2.2.3 Image of an extended object: Spherical mirror

$$m = \frac{y'}{y} = -\frac{s'}{s}$$

Lateral magnification

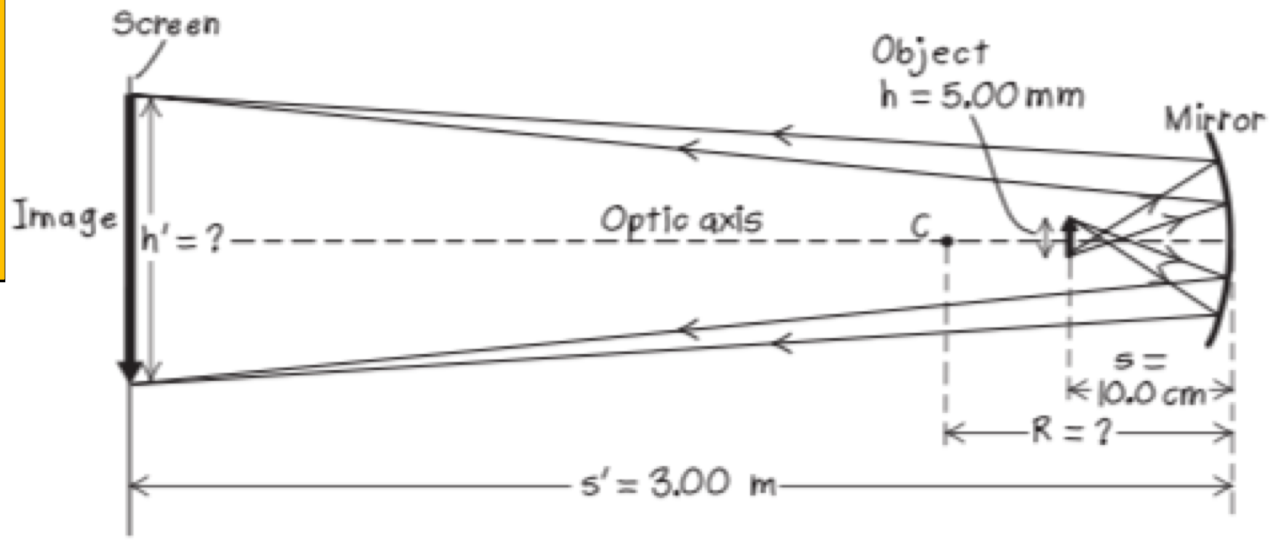


Exp.2.1: Image formation by a concave mirror I

A concave mirror forms an image, on a wall 3.00m in front of the mirror, of a headlamp filament 10.0cm in front of the mirror.

(a) What are the radius of curvature and focal length of the mirror?

(b) What is the lateral magnification? What is the image height if the object height is 5.00mm?

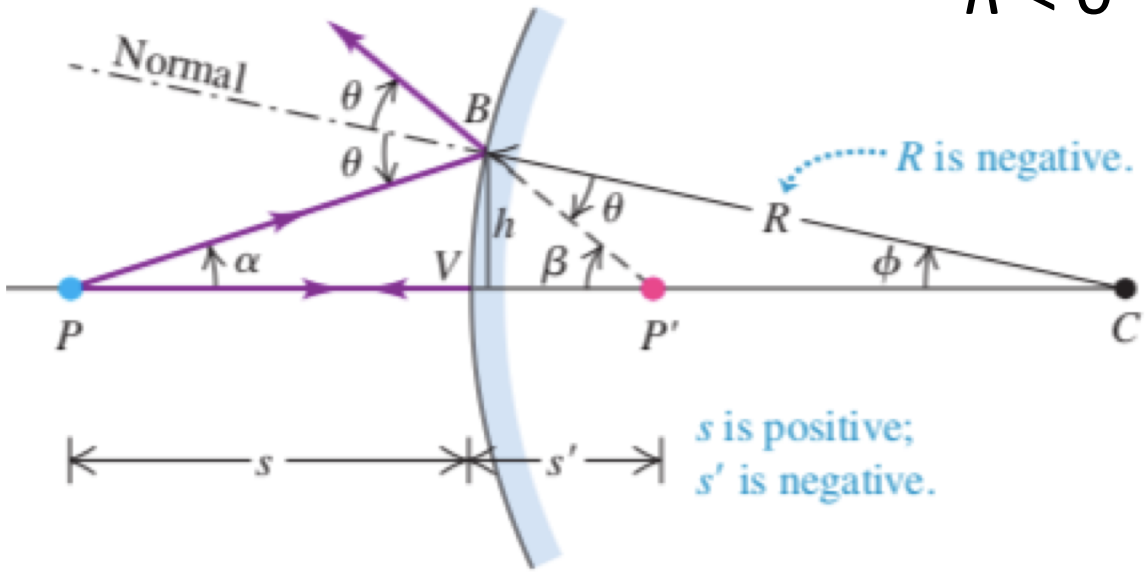


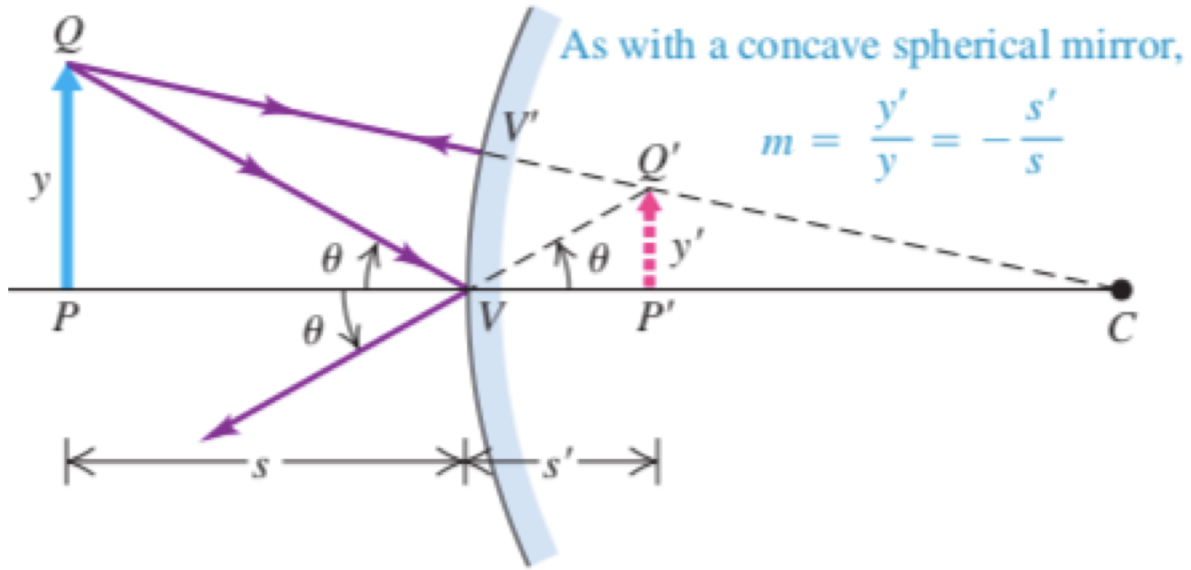
Chapter 2. Geometric optics (5.0h)

2.2 Reflection at a spherical surface

2.2.4 Convex mirrors

$R < 0$





$$\frac{1}{s} + \frac{1}{s'} = \frac{2}{R}$$

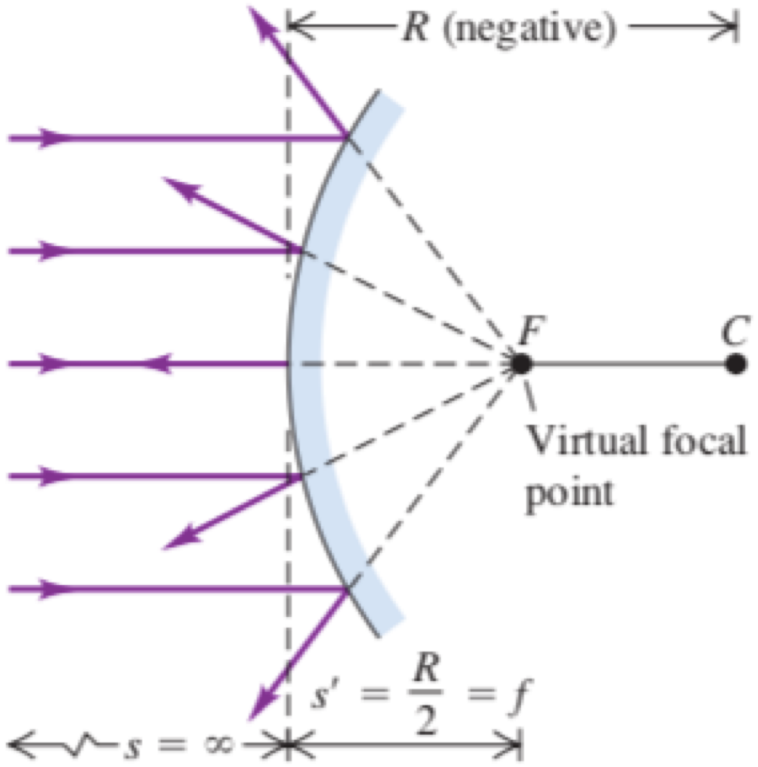
$$m = \frac{y'}{y} = -\frac{s'}{s}$$

Lateral magnification

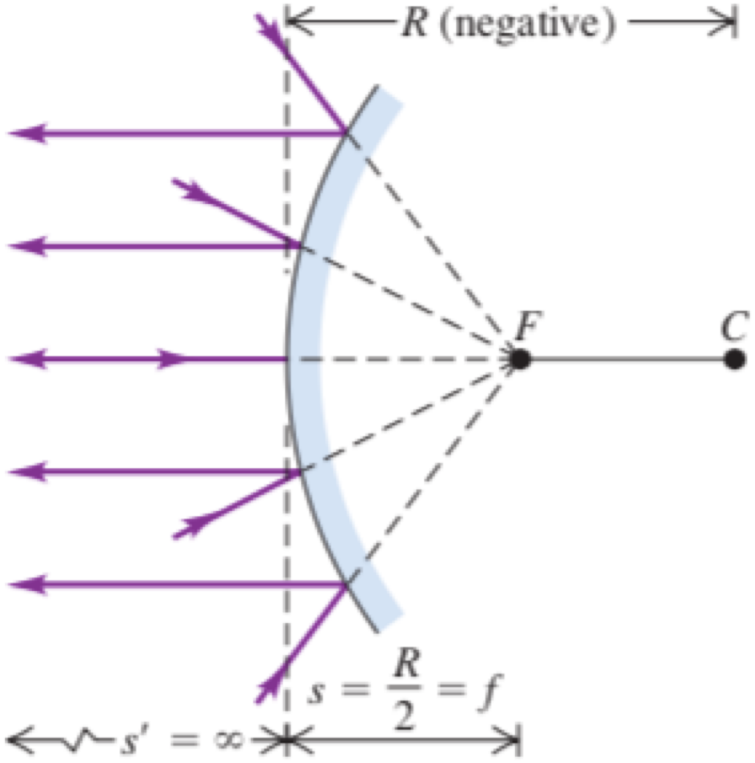
Chapter 2. Geometric optics (5.0h)

2.2 Reflection at a spherical surface

2.2.4 Convex mirrors



Virtual focal point *F*
(Behind the mirror)



Focal length $f = \frac{R}{2}$

$$\frac{1}{s} + \frac{1}{s'} = \frac{1}{f}$$

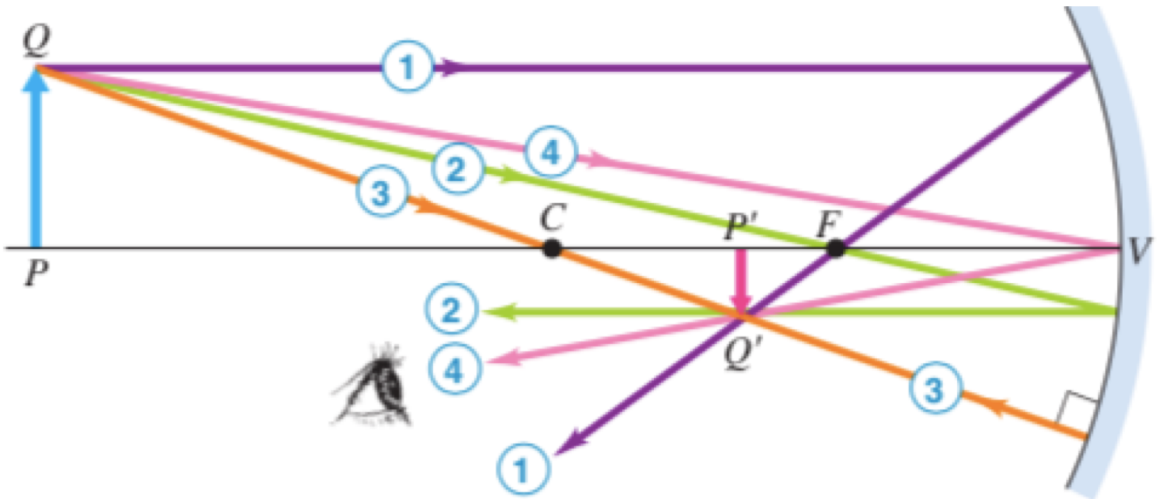
Chapter 2. Geometric optics (5.0h)

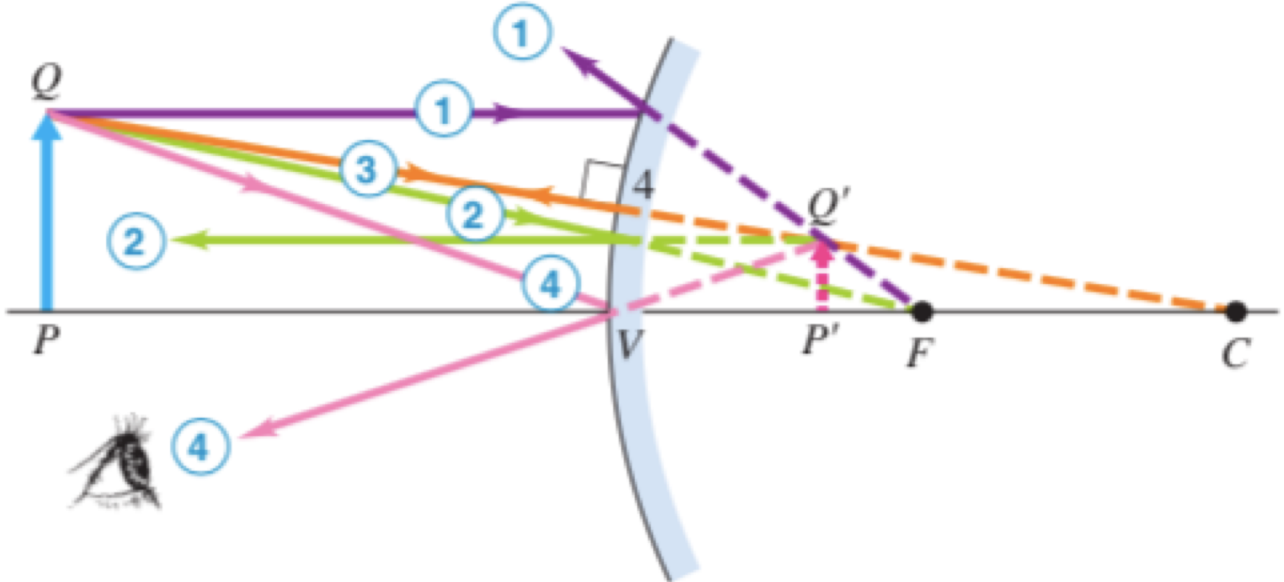
2.2 Reflection at a spherical surface

2.2.5 Graphical methods for mirrors

Principle rays

1. A ray **parallel to axis**, after reflection, passes through the focal point *F* of a concave mirror
   appears to come from the virtual focal point of a convex mirror
2. A ray **through (or proceeding toward)** the focal point F is reflected parallel to the axis
3. A ray **along the radius** through or away the center of curvature *C* intersects the surface normally and is reflected back along its original path.
4. A ray **to the vertex V** is reflected forming equal angles with the optic axis.



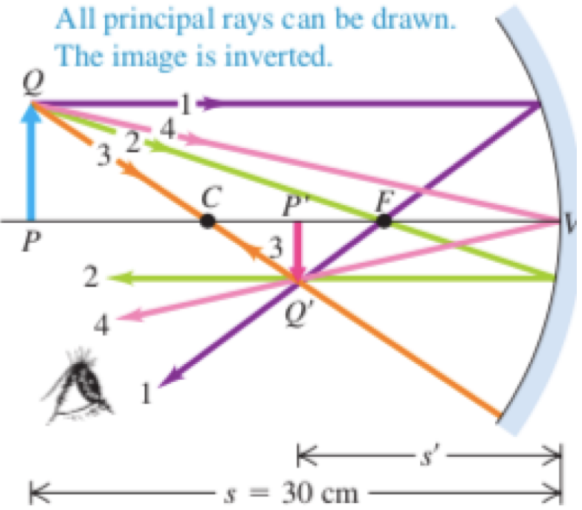


Chapter 2. Geometric optics (5.0h)

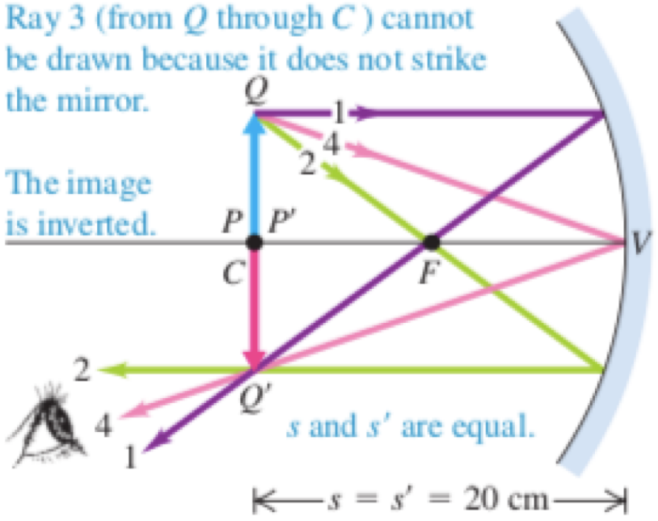
2.2 Reflection at a spherical surface

Exp.2.3: Concave mirror with various object distances

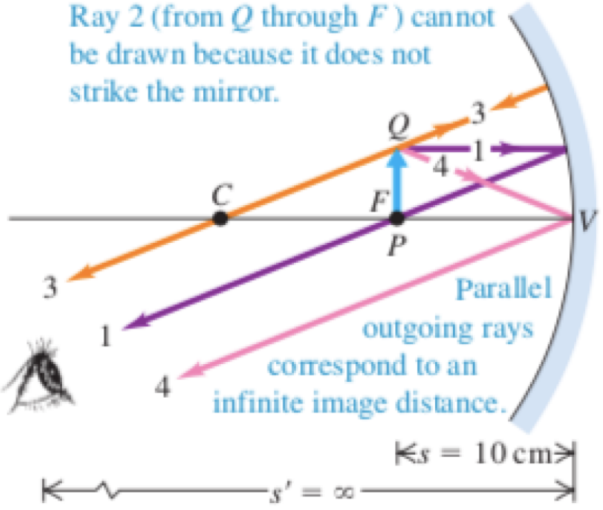
A concave mirror has a radius of curvature with absolute value 20 cm. Find graphically the image of an object in the form of an arrow perpendicular to the axis of the mirror at object distances of (a) 30 cm, (b) 20 cm, (c) 10 cm, and (d) 5 cm. Check the construction by *computing* the size and lateral magnification of each image.



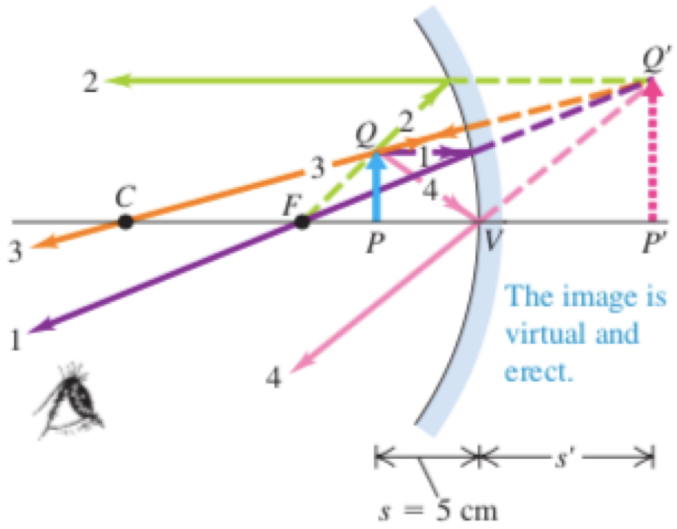
(a) d = 30 cm



(b) d = 20 cm



(c) d = 10 cm



(d) d = 5 cm

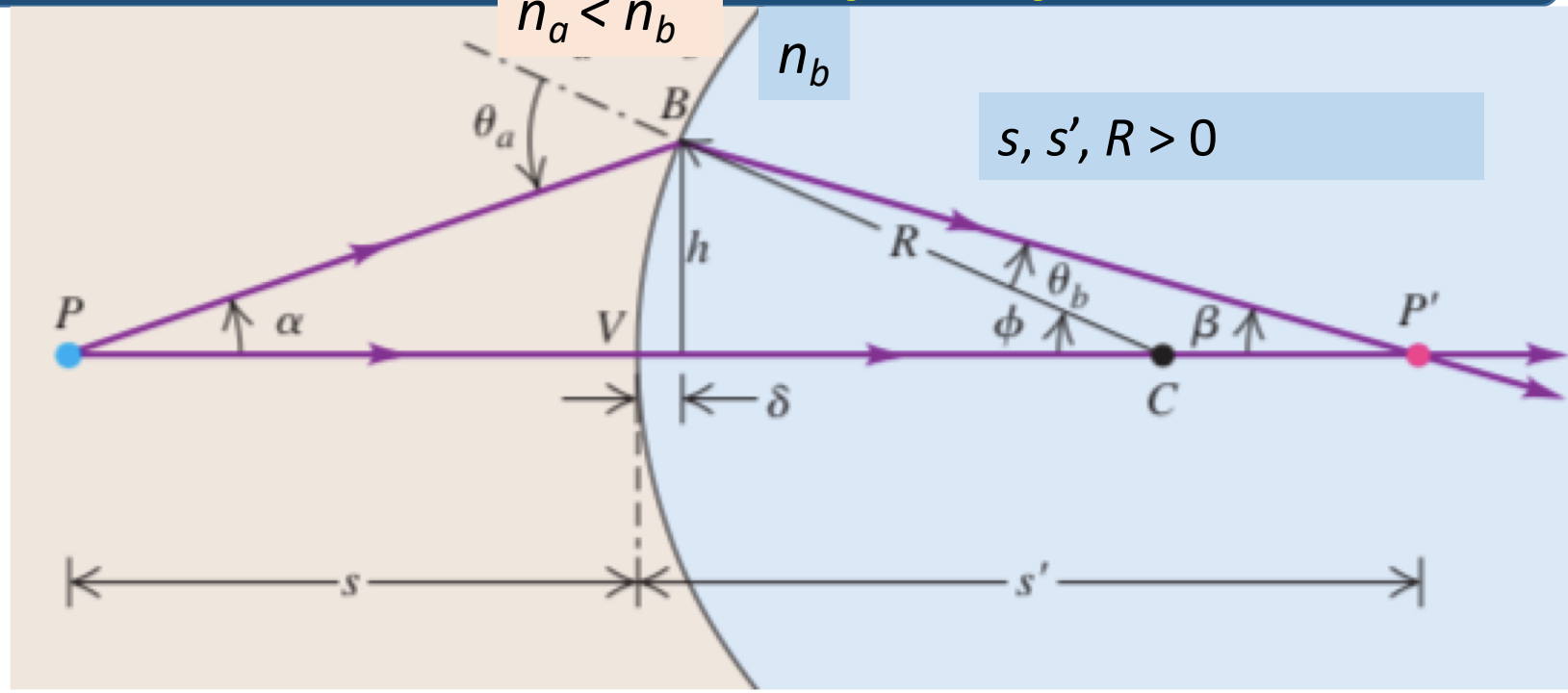
Chapter 2. Geometric optics (5.0h)

2.3 Refraction at a spherical surface

2.3.1 Image of a point object: Spherical Refracting surface

- A spherical surface with radius R
- Two materials with refractive indexes n_a and n_b
- P' is image of point P

s = ? s' = ?



The image P' of P formed by refraction at a spherical surface.
The refractive indexes of two materials are n_a and n_b ($n_a < n_b$)

- PV strikes the sphere

=> passes straight into the second material.

- PB is incident at an angle θ_a

=> is refracted at an angle θ_b

} intersect at P'

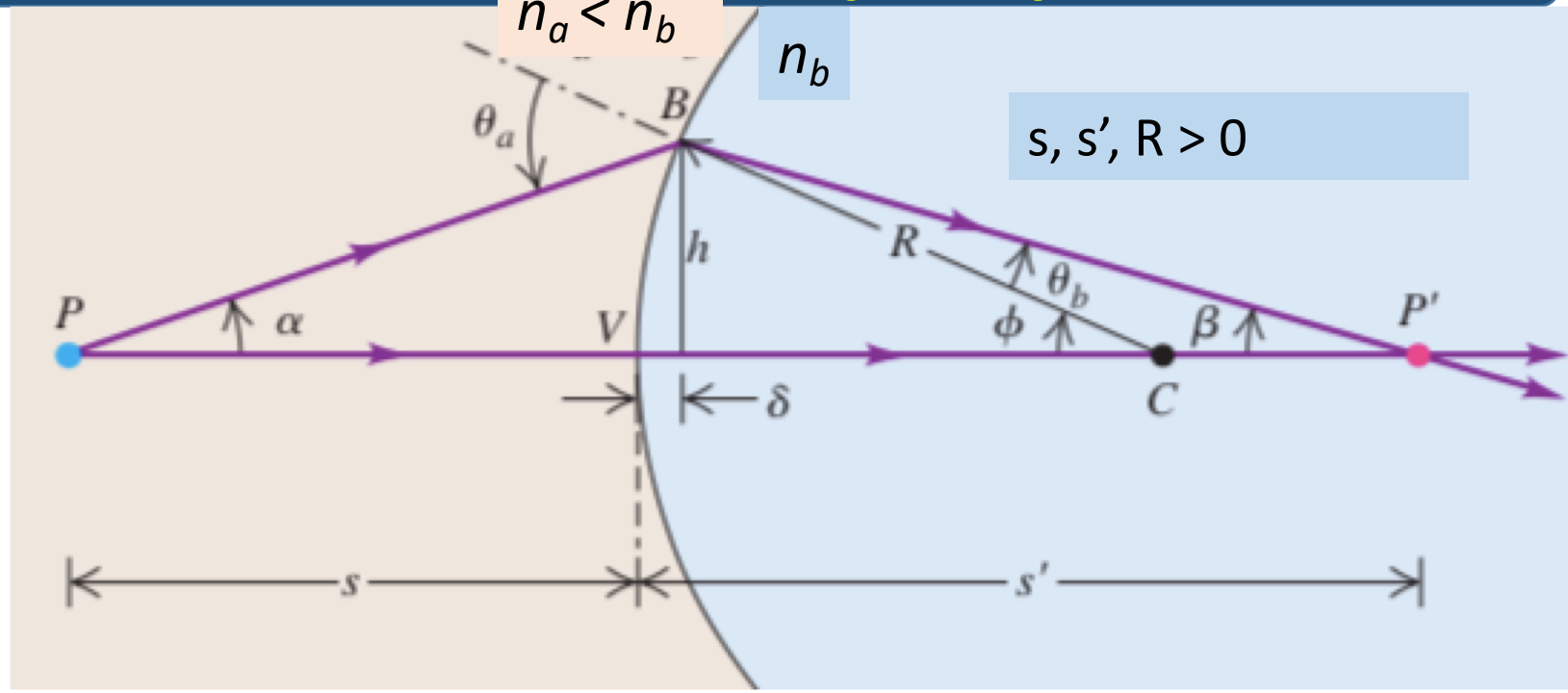
$s = PV$

$s' = P'V$

Chapter 2. Geometric optics (5.0h)

2.3 Refraction at a spherical surface

2.3.1 Image of a point object: Spherical Refracting surface



- Consider the triangles *PBC* and *P'BC*:

$$\theta_a = \alpha + \phi$$

$$\phi = \beta + \theta_b$$

- Snell's law:

$$n_a \sin\theta_a = n_b \sin\theta_b$$

- $\theta \ll$: $\sin\theta \approx \theta$

$$n_a\theta_a = n_b\theta_b \Rightarrow \theta_b = \frac{n_a}{n_b}(\alpha + \phi) \Rightarrow n_a\alpha + n_b\beta = (n_b - n_a)\phi$$

$$\tan\alpha = \frac{h}{s + \delta} \qquad \tan\beta = \frac{h}{s' - \delta} \qquad \tan\phi = \frac{h}{R - \delta}$$

- $\theta \ll$: $\tan\theta \approx \theta$ $(\delta \ll)$

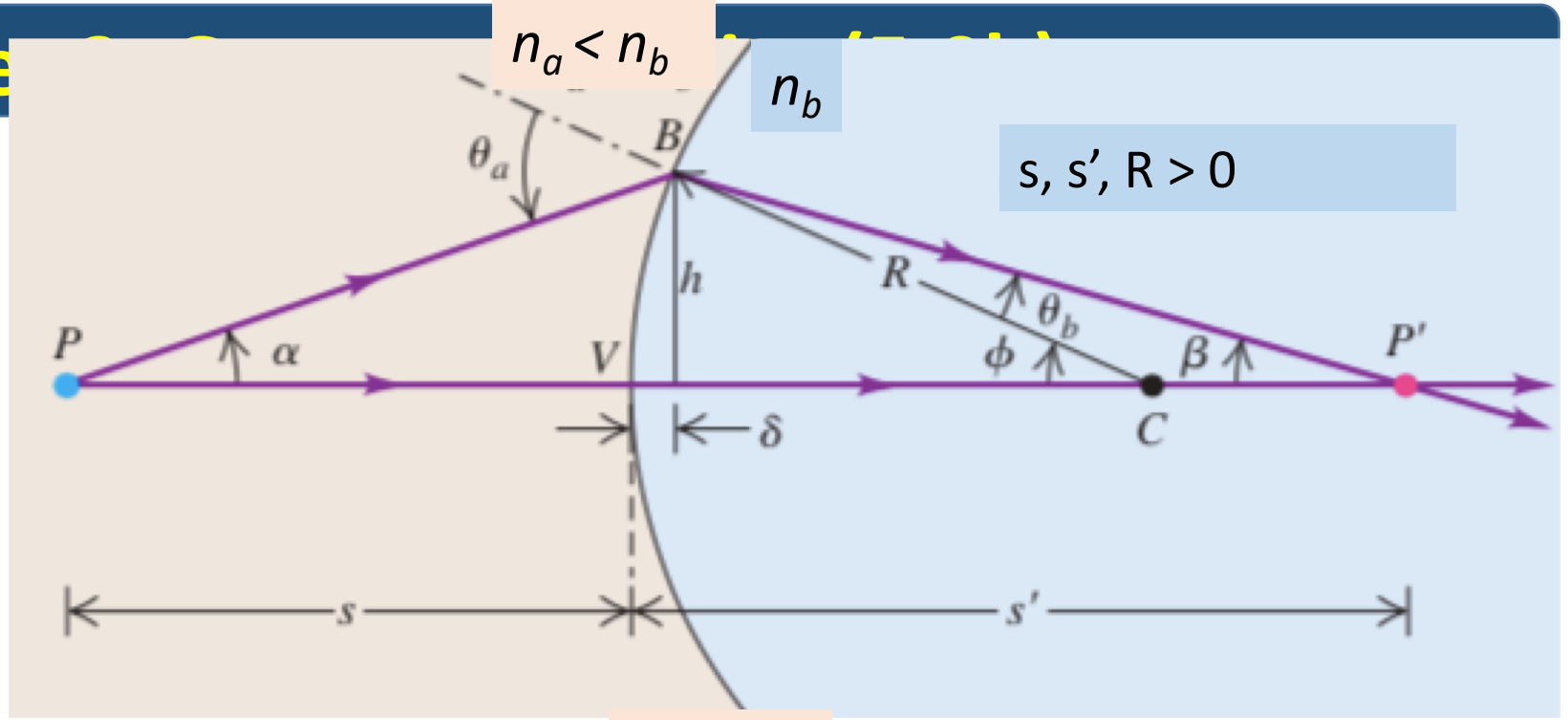
$$\alpha = \frac{h}{s} \qquad \beta = \frac{h}{s'} \qquad \phi = \frac{h}{R}$$

2.3 Refraction at a spherical surface

2.3.1 Image of a point object: Spherical Refracting surface

- Object-Image relationship, Spherical refracting surface

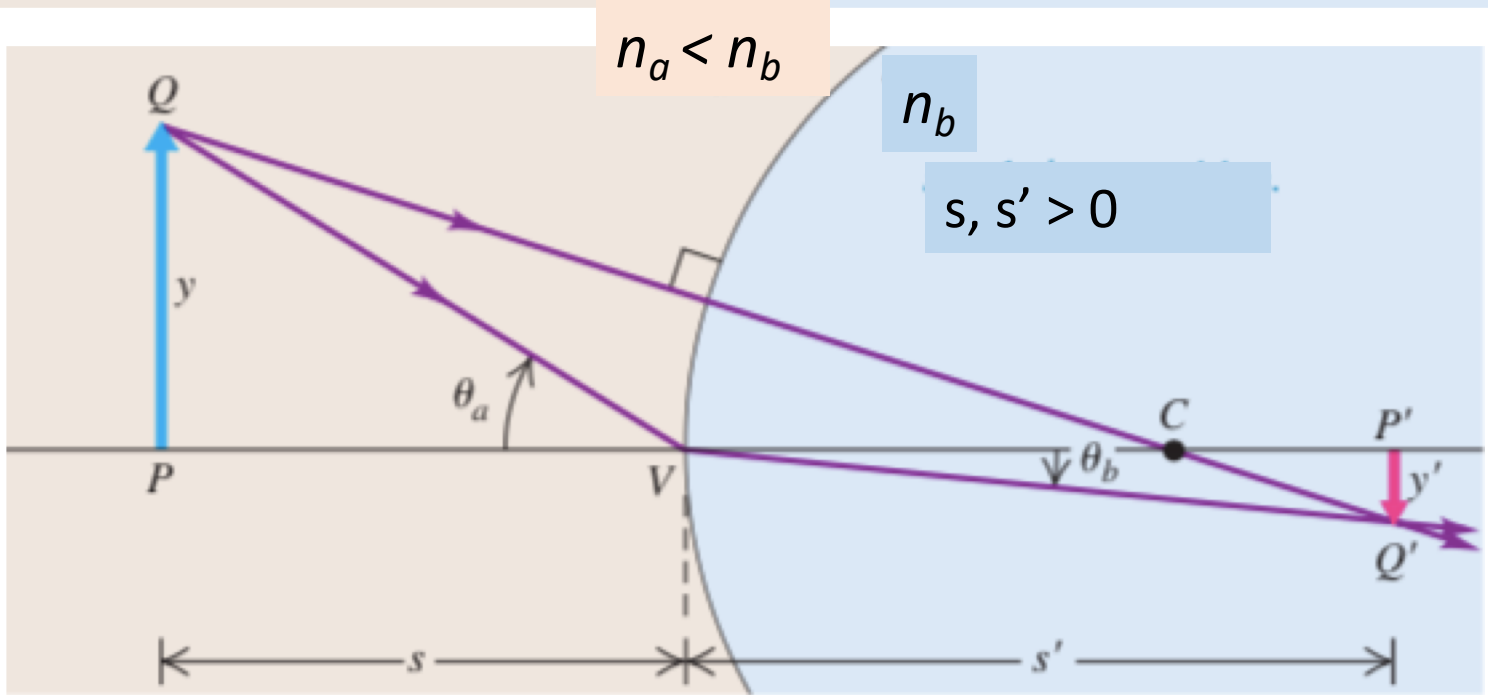
$$\frac{n_a}{s} + \frac{n_b}{s'} = \frac{n_b - n_a}{R}$$



2.3.2 Image of an extended object: Spherical Refracting surface

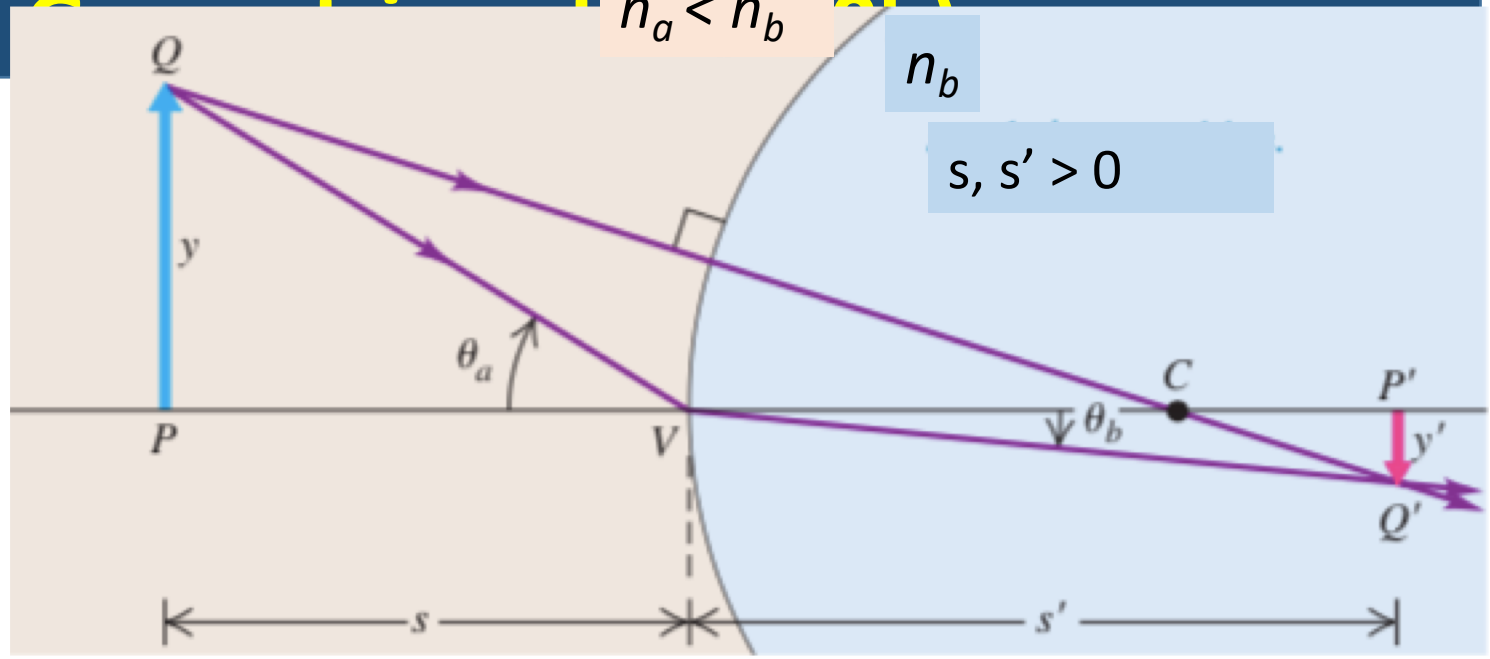
- Lateral magnification, Spherical refracting surface

$$m = \frac{y'}{y} = -\frac{n_a s'}{n_b s}$$



2.3 Refraction at a spherical surface

2.3.2 Image of an extended object: Spherical Refracting surface



An image formed by refraction at a spherical surface. The refractive indexes of two materials are n_a and n_b ($n_a < n_b$).

- *P'Q'* is image of an extended object *PQ*

- *QV* strikes the sphere at an angle θ_a

=> refracts at an angle θ_b

- *QC* is incident through *C*

=> refracts along *C*

} intersect at *Q'*

- Consider the triangles *PQV* and *P'Q'V*:

$$\tan\theta_a = \frac{y}{s} \qquad \tan\theta_b = \frac{-y'}{s'}$$

- Snell's law:

$$n_a \sin\theta_a = n_b \sin\theta_b$$

- $\theta \ll$:

$$\tan\theta \approx \sin\theta$$

$$\frac{n_a y}{s} = -\frac{n_b y'}{s'}$$

- Lateral magnification, Spherical refracting surface

$$m = \frac{y'}{y} = -\frac{n_a s'}{n_b s}$$

Chapter 2. Geometric optics (5.0h)

2.3 Refraction at a spherical surface

2.3.3 Image of an object: Plane refracting surface

- Object-Image relationship, Plane refracting surface (R=∞)

$$\frac{n_a}{s} + \frac{n_b}{s'} = 0$$

- Lateral magnification, Plane refracting surface

$$m = 1$$ The same lateral size, erect.

Chapter 2. Geometric optics (5.0h)

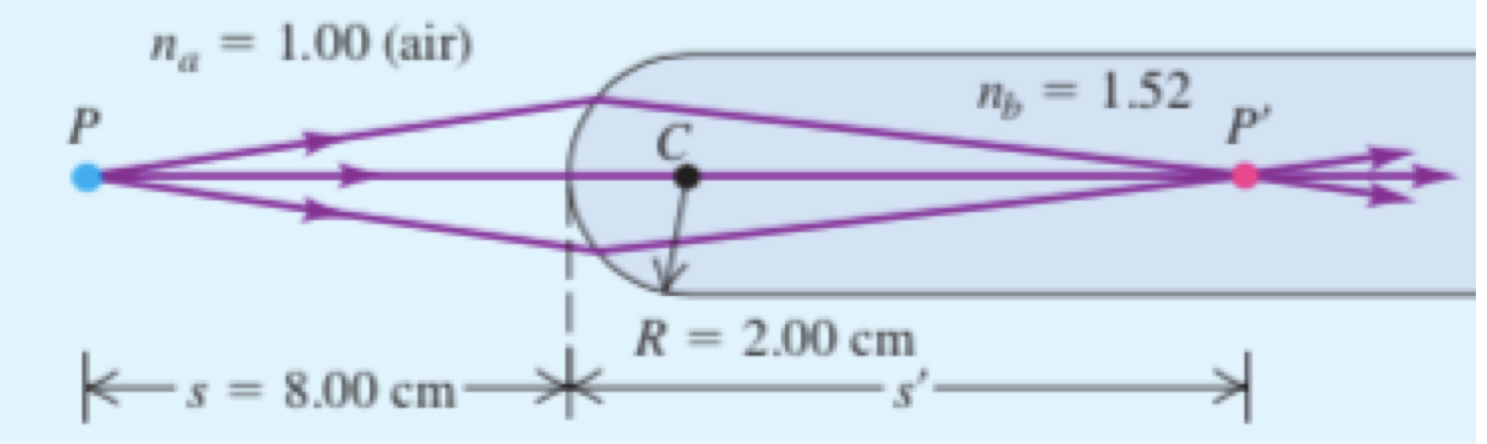
2.3 Refraction at a spherical surface

Exp.2.4: Image formation by refraction

A cylindrical glass rod has index of refraction 1.52. It is surrounded by air. One end is ground to a hemispherical surface with radius *R* = 2.00 cm. A small object is placed on the axis of the rod, 8.00 cm to the left of the vertex. Find (a) the image distance and (b) the lateral magnification.

$$\frac{n_a}{s}+\frac{n_b}{s'}=\frac{n_b-n_a}{R}$$

$$m=\frac{y'}{y}=-\frac{n_a s'}{n_b s}$$



(a) The image distance $\frac{1}{8}+\frac{1.52}{s'}=\frac{1.52-1}{2}$ => s' = 11.3cm

(b) The lateral magnification $m=-\frac{1\times 11.3}{1.52\times 8}=-0.929$

Chapter 2. Geometric optics (5.0h)

2.4 Thin lenses

- A lens is an optical system with two refracting surfaces.
- A thin lens: has two spherical surfaces close enough together that the distance between them (the thickness of the lens) can be neglected.

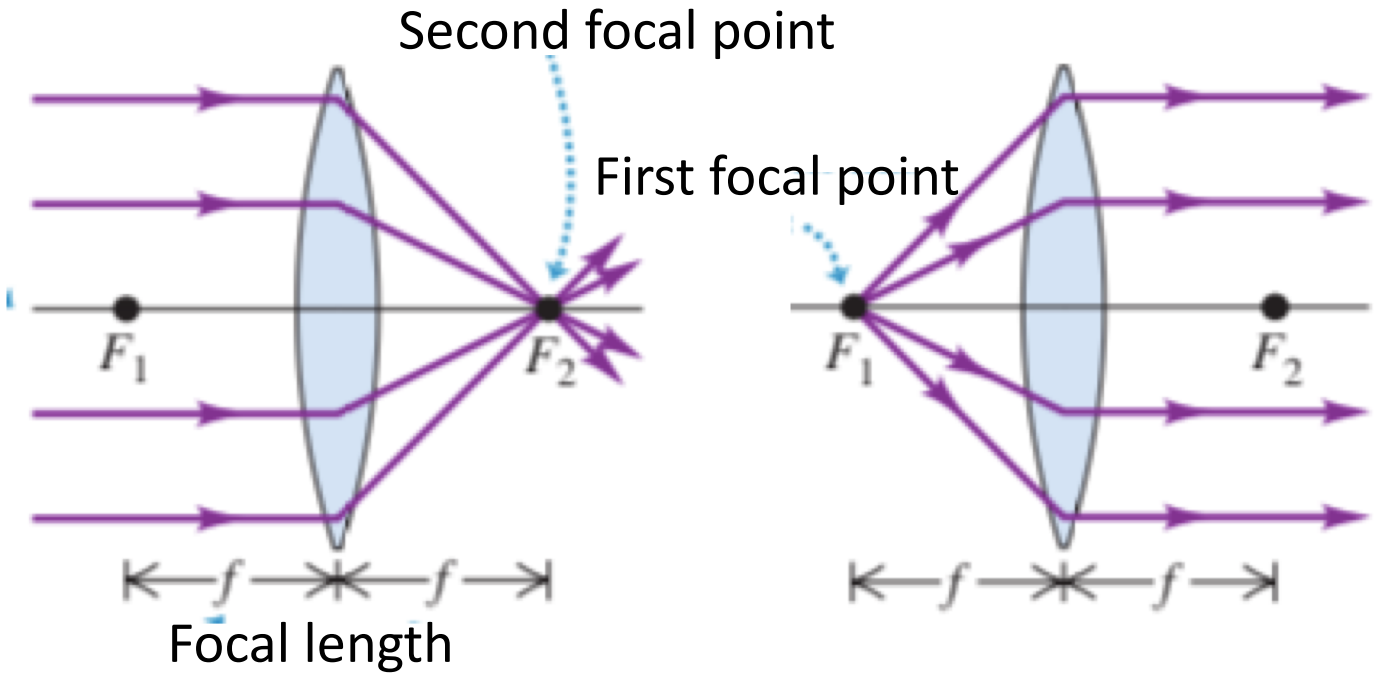
2.4.1 Properties of a lens

- **A converging thin lens**:

✓ a beam of rays // to the axis passes through the lens,
=> the rays converge to a point F_2.

✓ rays passing through F_1
=> emerge from the lens as a beam of // rays.

- F_1, F_2: first and second **focal points**.
- f: distance from the focal point to the center of the lens: **focal length**.
- The line connecting the centers of curvature of two spherical surfaces: **optic axis**.



A converging thin lens.
$f > 0$

Chapter 2. Geometric optics (5.0h)

2.4 Thin lenses

2.4.2 Image of an extended object: Converging lens

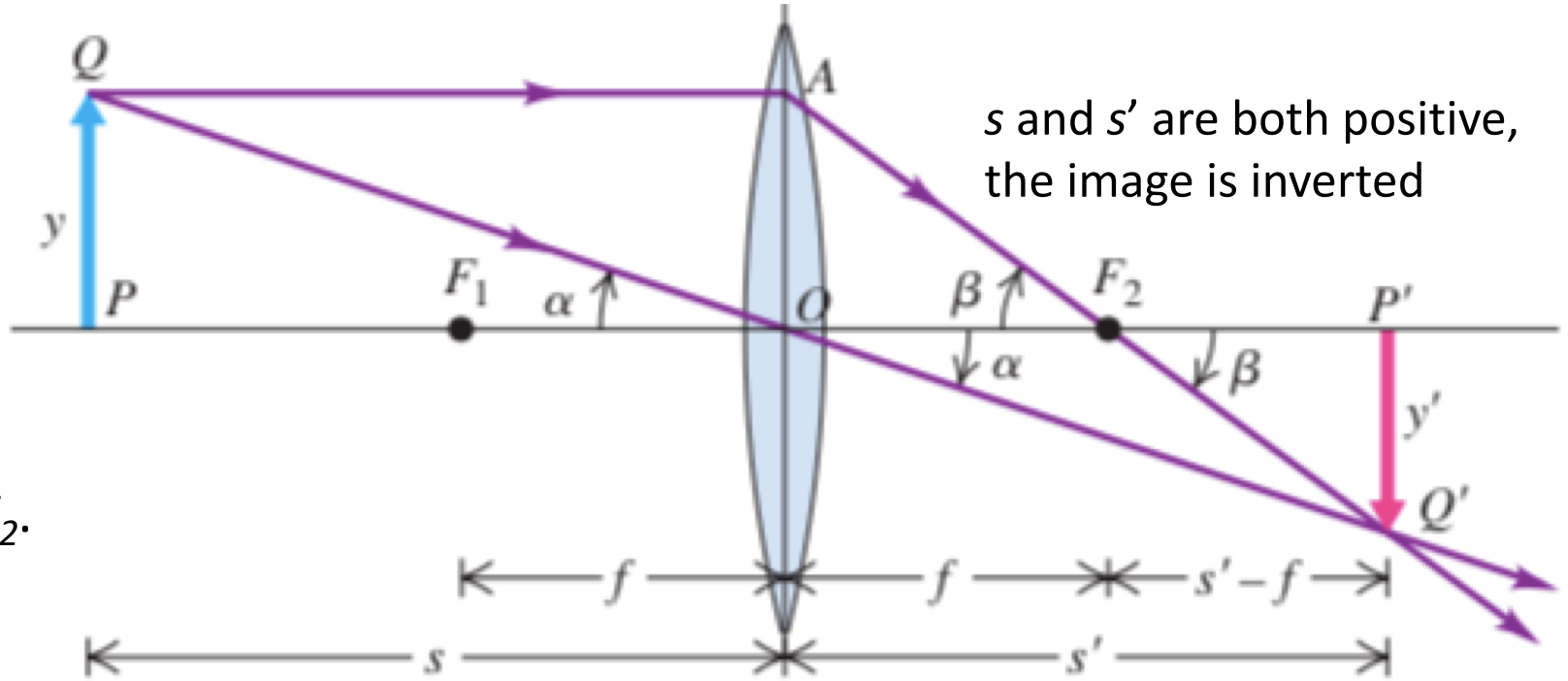
s, *s'*: the object and image distances

y, *y'*: the object and image heights

- *QA*: // the optic axis

=> passes through the 2nd focal point F_2.

- *QOQ'*: passes undeflected straight through the center of the lens



s and *s'* are both positive, the image is inverted

Construction used to find image position for a thin lens

- Consider the similar triangles *PQO* and *P'Q'O*:

$$\frac{y'}{y} = -\frac{s'}{s}$$

- Consider the similar triangles OAF_2 and $P'Q'F_2$:

$$\frac{y'}{y} = -\frac{s' - f}{f}$$

$$\frac{1}{s} + \frac{1}{s'} = \frac{1}{f}$$ Object-Image relationship, thin lens

$$m = \frac{y'}{y} = -\frac{s'}{s}$$ Lateral magnification, thin lens

Chapter 2. Geometric optics (5.0h)

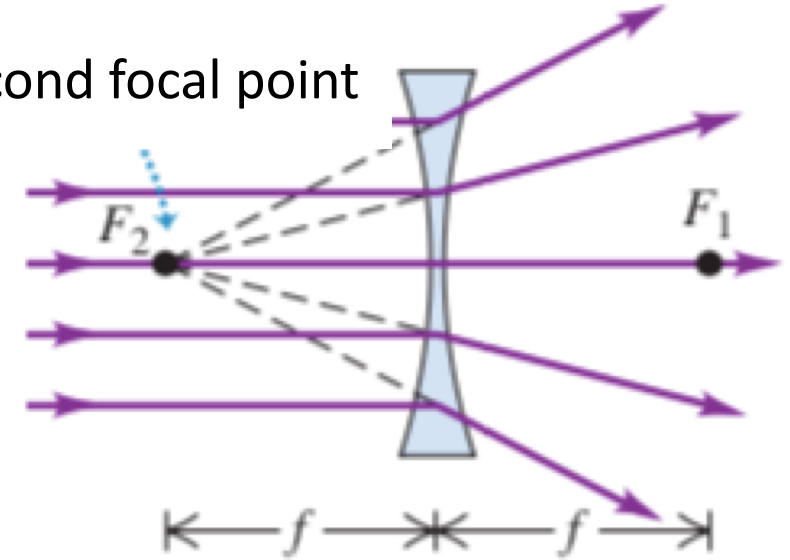
2.4 Thin lenses

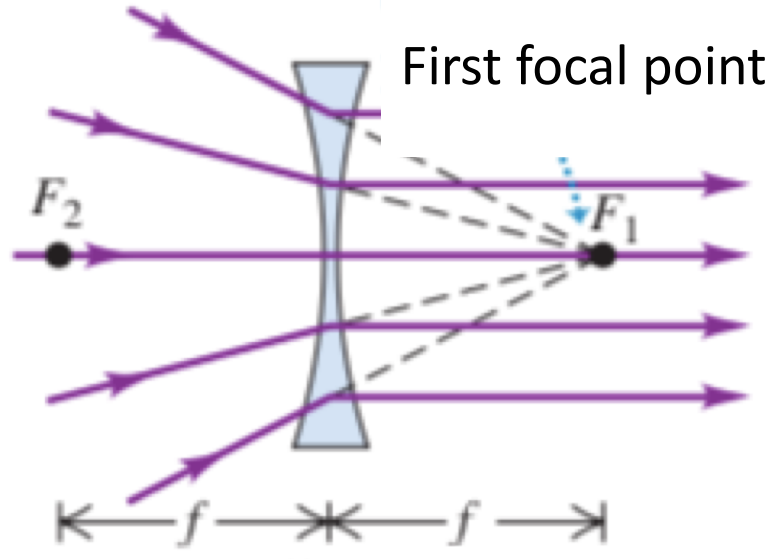
2.4.3 Diverging lens

- **A diverging thin lens**:

✓ a beam of rays // to the axis passes through the lens,
=> the rays diverge to a point F_2.

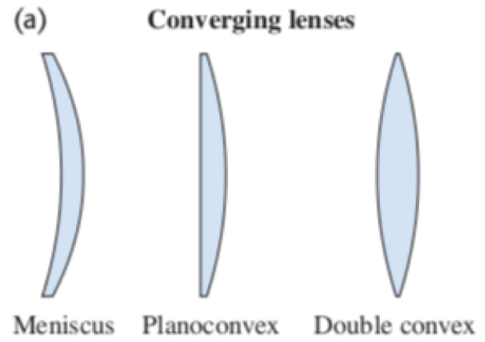
✓ incident rays converging toward the first focal point F_1
=> emerge from the lens // to axis.

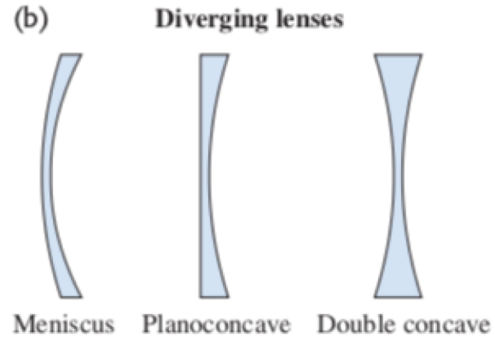




A diverging thin lens.
$f < 0$

- Any lens that is thicker at its center than at its edges is a **converging lens** with **positive** *f*;
- Any lens that is thicker at its edges than at its center is a **diverging lens** with **negative** *f*

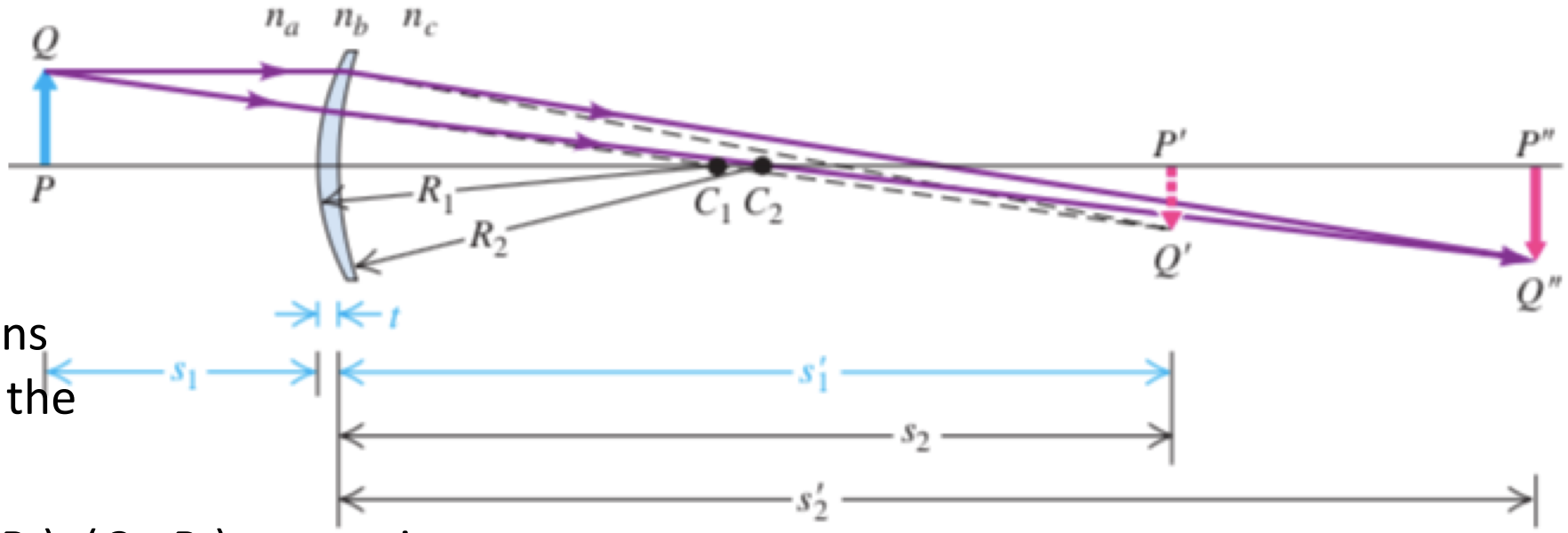




Chapter 2. Geometric optics (5.0h)

2.4 Thin lenses

2.4.4 The lensmaker's equation



- The relationship among:
  - ✓ the focal length *f*,
  - ✓ the refractive index *n* of the lens
  - ✓ the radii of curvature R_1, R_2 of the lens surfaces.

- Two spherical interfaces (C_1, R_1), (C_2, R_2) separating three materials with refractive indexes n_a, n_b, and n_c
- s_1, s_1': the object and image distances for the 1st surface
- s_2, s_2': the object and image distances for the 2nd surface

- $t \ll s_1, s_{1'}, s_2, s_2'$

=> $s_2 = -s_1'$

$$\frac{n_a}{s_1}+\frac{n_b}{s_1'}=\frac{n_b-n_a}{R_1}$$

$$\frac{n_b}{s_2}+\frac{n_c}{s_2'}=\frac{n_c-n_b}{R_2}$$

$n_a = n_c = 1$, $n_b = n$, $s_2 = -s_1'$

$$\frac{1}{s_1}+\frac{n}{s_1'}=\frac{n-1}{R_1}$$

$$-\frac{n}{s_1'}+\frac{1}{s_2'}=\frac{1-n}{R_2}$$

$$\frac{1}{s_1}+\frac{1}{s_2'}=(n-1)\left(\frac{1}{R_1}-\frac{1}{R_2}\right)$$

$$\frac{1}{s}+\frac{1}{s'}=(n-1)\left(\frac{1}{R_1}-\frac{1}{R_2}\right)$$

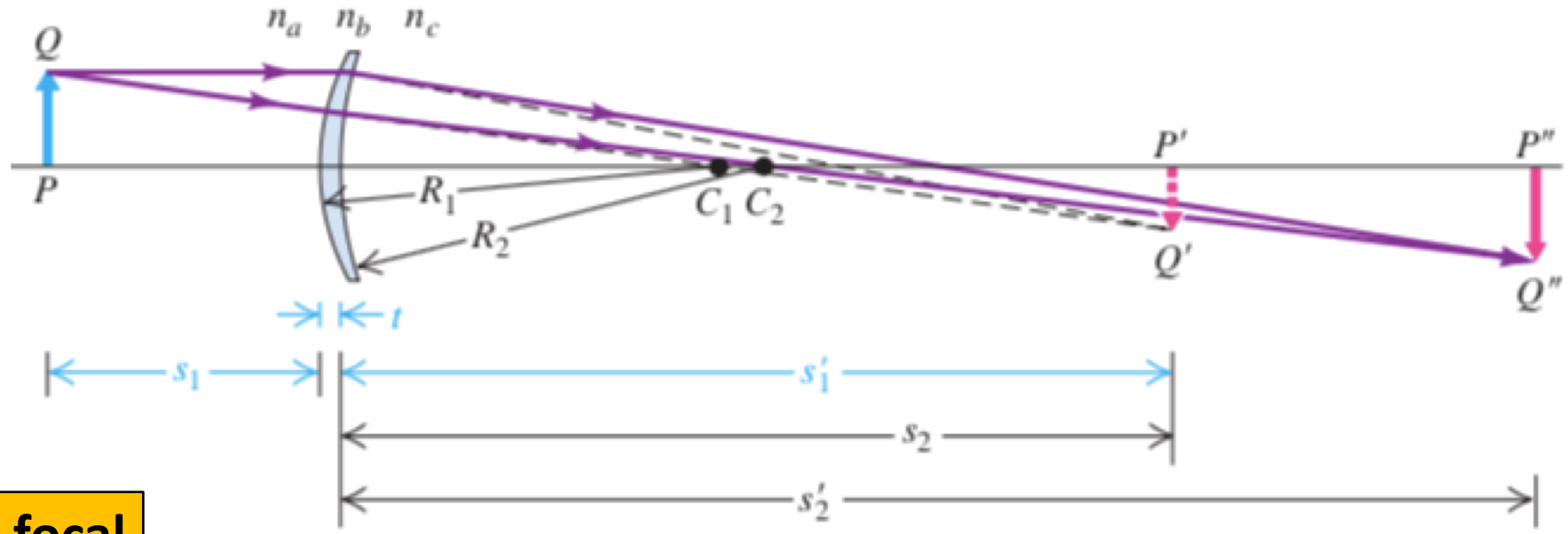
Chapter 2. Geometric optics (5.0h)

2.4 Thin lenses

2.4.4 The lensmaker's equation

$$\frac{1}{f} = (n-1)\left(\frac{1}{R_1} - \frac{1}{R_2}\right)$$

lensmaker equation for a thin lens

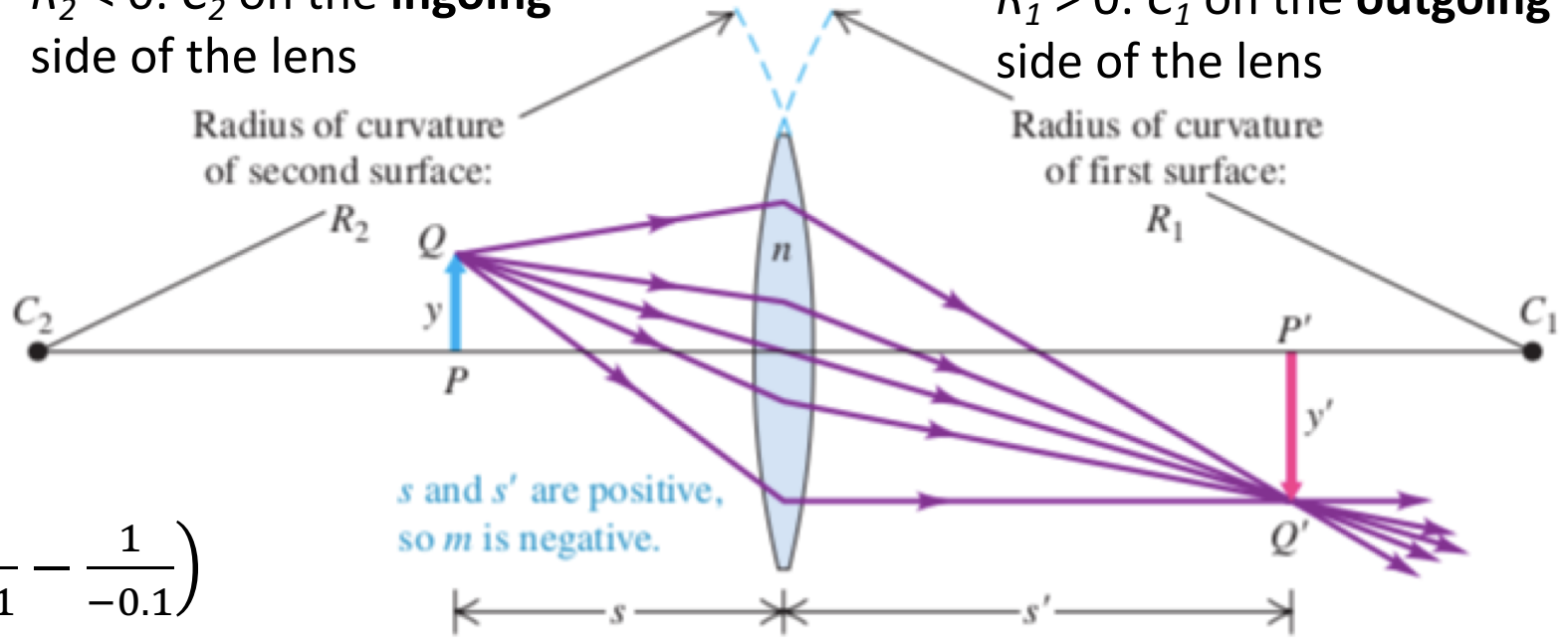


Exp.2.5: Determining the focal length of a lens

Suppose the absolute value of the radii of curvature of the lens surface in this following figure are both equal to 10 cm and the refractive index of the glass is *n* = 1.52. What is the focal length *f* of the lens?

$R_2 < 0$: C_2 on the **ingoing** side of the lens

$R_1 > 0$: C_1 on the **outgoing** side of the lens



$$\frac{1}{f} = (n-1)\left(\frac{1}{R_1} - \frac{1}{R_2}\right) = (1.52-1)\left(\frac{1}{0.1} - \frac{1}{-0.1}\right)$$

Chapter 2. Geometric optics (5.0h)

2.4 Thin lenses

2.4.5 Graphical methods for lenses

1. A ray **parallel to the axis** emerges from the lens in a direction that

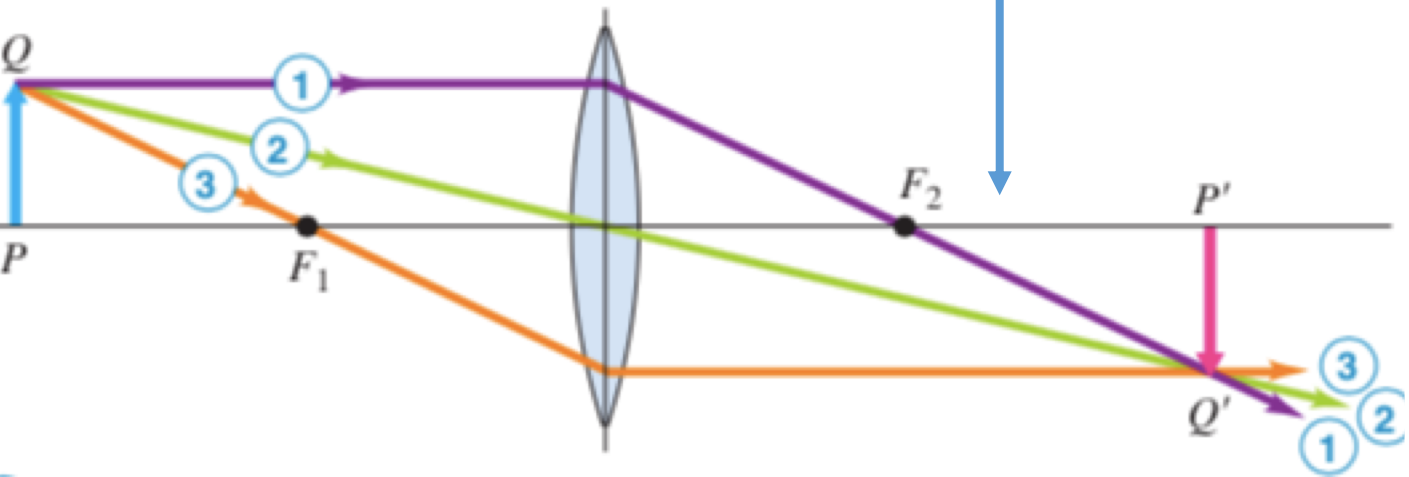
passes through the second focal point F_2 of a **converging lens**

appears to come from the second focal point of a **diverging lens**

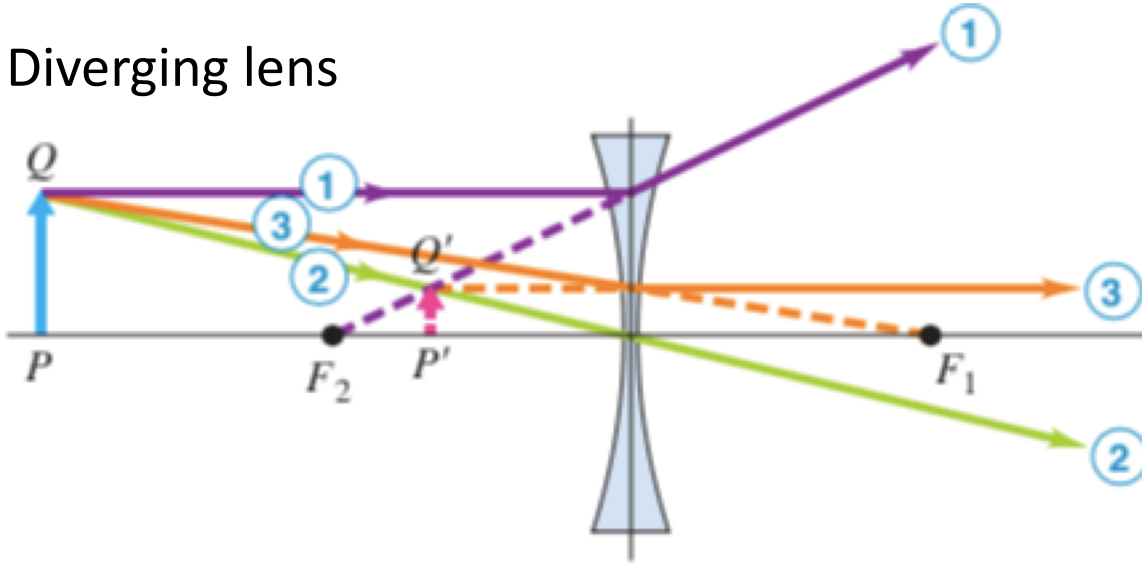
2. A ray **through the center of the lens** emerges at essentially the same angle at which it enters and along essentially the same line.

3. A ray **through (proceeding toward) the first focal point *F*$_1$** emerges parallel to the axis.

Converging lens



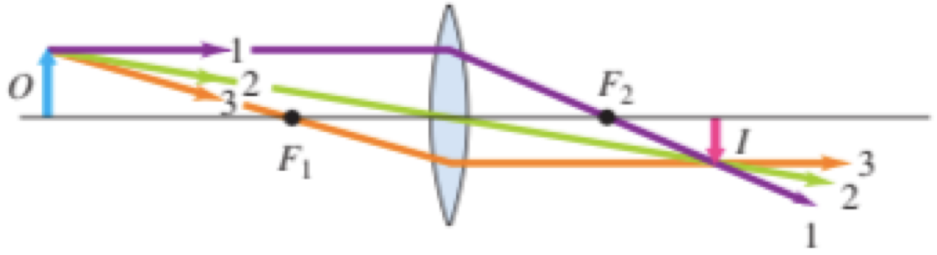
Diverging lens



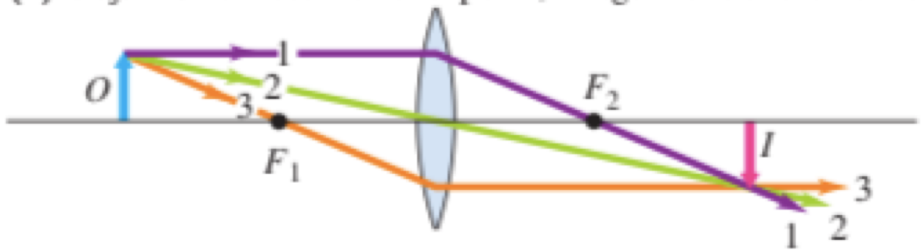
Chapter 2. Geometric optics (5.0h)

2.4 Thin lenses

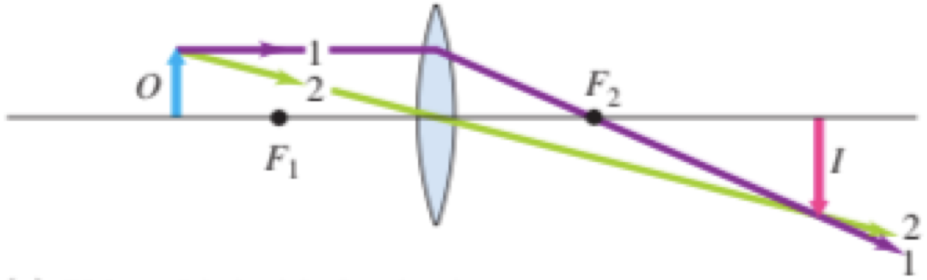
(a) Object O is outside focal point; image I is real.



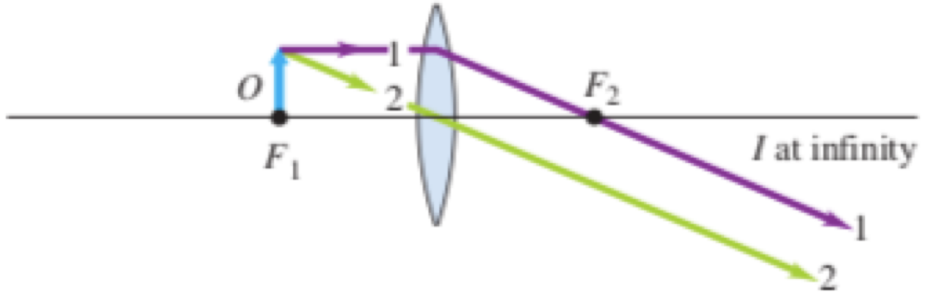
(b) Object O is closer to focal point; image I is real and farther away.



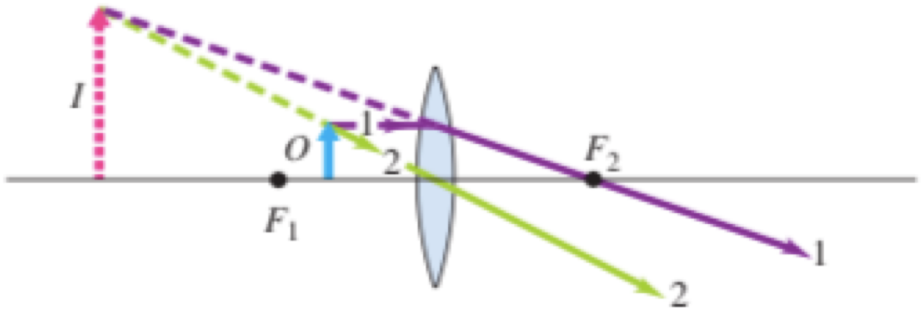
(c) Object O is even closer to focal point; image I is real and even farther away.



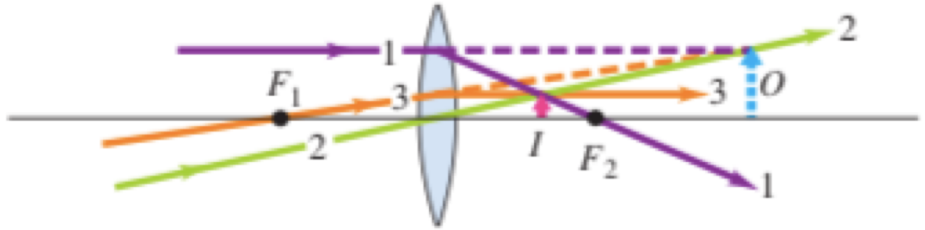
(d) Object O is at focal point; image I is at infinity.



(e) Object O is inside focal point; image I is virtual and larger than object.



(f) A virtual object O (light rays are *converging* on lens)



2.4 Thin lenses

Exp.2.6: Image position and magnification by a converging lens
Find the image position and magnification for an object at each of the following distances from a converging lens with a focal length of 20 cm: (a) 50 cm; (b) 20 cm; (c) 15 cm; (d) -40 cm. Draw the rays to find the image and mention the properties of the image.

$\frac{1}{s} + \frac{1}{s'} = \frac{1}{f}$ Object-Image relationship, thin lens

$m = -\frac{s'}{s}$ Lateral magnification, thin lens

f = 20 cm

(a) s = 50 cm => s' = 33.3 cm; m = -2/3
(b) s = 20 cm => s' = ±∞ ; m = ±∞
(c) s = 15 cm => s' = -60 cm ; m = 4
(d) s = -40 cm => s' = 13.3 cm ; m = 1/3

Chapter 2. Geometric optics (5.0h)

2.4 Thin lenses

Exp.2.7: Image position and magnification by a diverging lens

A beam of parallel rays spreads out after passing through a thin diverging lens, as if the rays all came from a point 20.0 cm from the center of the lens. You want to use this lens to form an erect, virtual image that is 1/3 the height of the object.

(a) Where should the object be placed? Where will the image be?

(b) Draw a principal ray diagram.

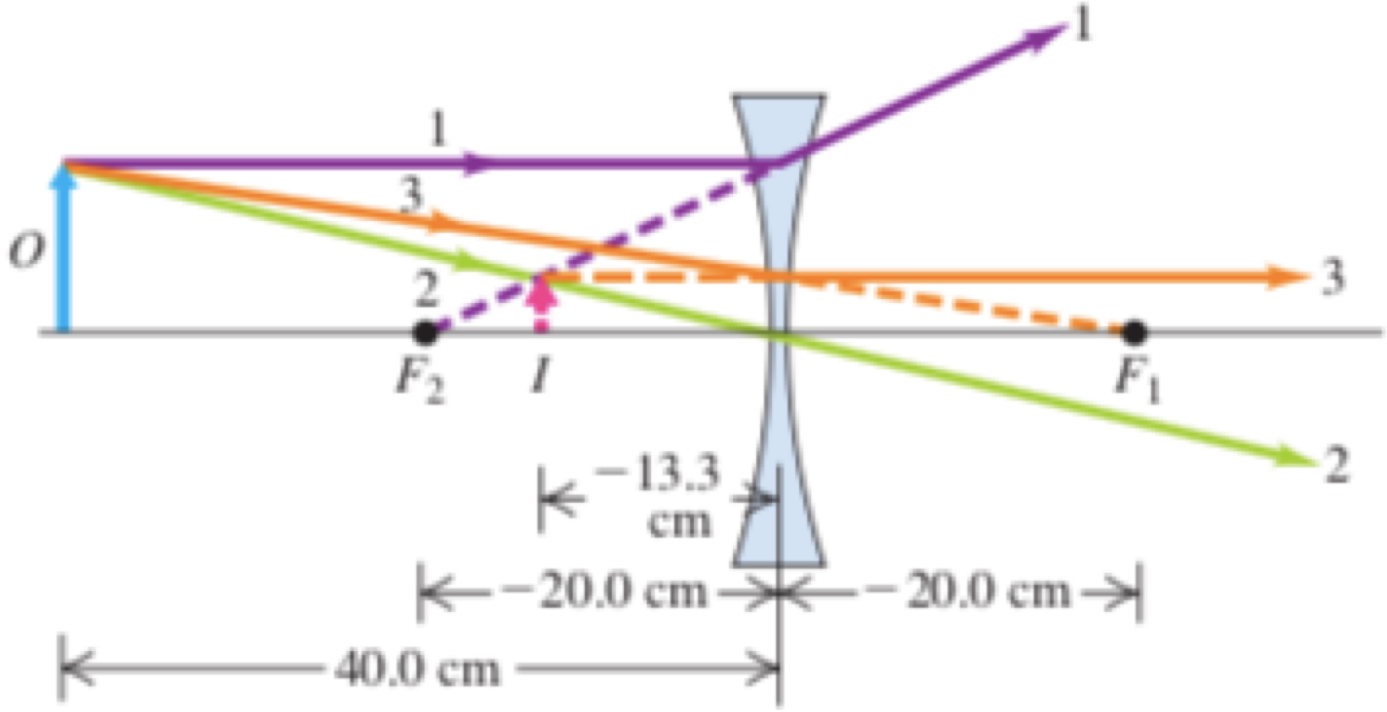
$$m = \frac{y'}{y} = -\frac{s'}{s} = \frac{1}{3}$$

s = 40 cm

s' = -13.3 cm

$$f = -20cm$$

$$\frac{1}{s} + \frac{1}{s'} = \frac{1}{f}$$



“When you really want something the whole universe conspires in helping you to achieve it”
(The Alchemist, Paulo Coelho)

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GOOD LUCK and THANK YOU

Make it Simple but Significant!